SPATIAL ANALYSIS AND MULTI-SOURCE DATA FUSION FOR DENGUE OUTBREAK FORECASTING USING AN ENSEMBLE OF RANDOM FOREST AND XGBOOST WITH HEATMAP VISUALIZATION

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Bachelor of Science in Computer Science
Specialized in Intelligent System

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APPROVAL SHEET

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DEDICATION

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ABSTRACT

Dengue remains a major public health concern in the Philippines, especially in Luzon, where outbreaks continue to affect communities each year. This study develops a forecasting approach that integrates epidemiological, demographic, meteorological, and spatial analysis to support proactive dengue management. Random Forest and XGBoost models were trained using multi-source data capturing disease patterns, climate variability, population-related risk factors and spatial analysis result. XGBoost produced stronger predictive accuracy compared to Random Forest, achieving an R-Squared of 0.926351, MAE of 4.628130, and RMSE of 7.802892. To further enhance prediction reliability, a Weighted Ensemble method was implemented with optimized weights of 0.1121 for Random Forest and 0.8879 for XGBoost. The ensemble model achieved the highest accuracy, resulting in an improved R-Squared of 0.926423, a lower MAE of 4.609461, and a reduced RMSE of 7.799064. A color-coded heatmap was used to visualize predicted dengue cases across localities and months, helping identify areas with higher outbreak risk. Therefore, the integration of machine learning and spatial-environmental variables strengthens forecasting performance and supports timely public health interventions for dengue control in Luzon.

Keywords: *Spatial Analysis, Multi-Source Data Fusion, Weighted Average Ensemble, Random Forest, XGBoost, Dengue Outbreak Forecasting*

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DEFINITION OF TERMS

Clustering	Clustering is an unsupervised machine learning technique used to identify and group similar data points together based on their characteristics or attributes. The goal of clustering is to partition a dataset into groups, called clusters, in such a way that the data points within a single cluster are highly similar to one another.
Coefficient of Determination (R^2)	The Coefficient of Determination (R2), or R-squared, measures how well the regression predictions approximate the actual data values. It quantifies the proportion of variance in the dependent variable that is explained by the independent variables in the model.
Dataset	A dataset is a single, organized collection of related data. It is the foundational unit for any data analysis, machine learning project, or research study. The data within the collection is grouped together because it pertains to a specific subject or project and is arranged in a specific format to make it accessible for processing and analysis.
DBSCAN	DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is an unsupervised clustering algorithm specifically designed to discover clusters of varying shapes and sizes in a dataset. In this study, DBSCAN was applied to perform spatial analysis and identify patterns and high-risk areas of dengue cases.

Demographic Data	This refers to population-related information such as population number and population density. These characteristics are important in dengue forecasting because densely populated areas may be more vulnerable to the spread of the disease. In this study, demographic data helps identify which communities are more at risk.
Dengue Outbreak	A dengue outbreak is a sudden increase in dengue cases in a specific place and time. It signals unusually high transmission and often requires urgent public health response.
Epidemiological Data	Epidemiological data are records about the occurrence of diseases, particularly the number of dengue cases over time. These data serve as the foundation for analyzing and predicting outbreak trends.
Feature Engineering	Feature Engineering is the process of extracting meaningful temporal and statistical features from raw data to capture the dynamic patterns of dengue transmission. In this study, this phase was implemented to create a structured and information-rich dataset suitable for forecasting.
Forecasting	Forecasting means predicting future events or conditions based on past and current information. In this research, it refers to estimating the likelihood of dengue outbreaks in advance using data-driven methods.

Getis Ord G*	The Getis-Ord Gi* statistic is a crucial spatial statistics tool used for hot spot analysis. This method calculates a z-score and a p-value for each feature in a dataset to determine if high-value (hotspots) or low-value (coldspots) clusters are statistically significant.
Hyperparameter Tuning	Hyperparameter Tuning is the automated process of systematically identifying the most effective combination of hyperparameters for a machine learning model to improve its performance. In this study, the models were optimized using Bayesian optimization through the Optuna framework to identify their optimal hyperparameter configurations.
Mean Absolute Error (MAE)	Mean Absolute Error (MAE) quantifies the average magnitude of prediction errors in a model. It is calculated by taking the absolute value of the difference between each predicted value and the actual value, then averaging those differences across all data points. A lower MAE indicates better model performance.
Meteorological Data	Meteorological data includes weather-related information such as rainfall, temperature, humidity etc. These factors affect mosquito breeding and survival, making them important for predicting dengue risks.
Multi-Source Data Fusion	Multi-source data fusion is the process of combining information from different datasets, such as weather, health, and population records.

Random Forest	Random Forest is a machine learning algorithm that builds many decision trees and combines their results for more accurate predictions. In this study, it is used as base model for weighted average ensemble.
Regression	Regression is a statistical method used to study the relationship between variables. Here, it helps estimate future dengue cases based on factors like past dengue records, weather, and population characteristics.
Root Mean Squared Error (RMSE)	Root Mean Squared Error (RMSE) measures the average magnitude of prediction errors, with larger errors receiving more weight due to squaring. It is calculated as the square root of the average of the squared differences between actual and predicted values. Because it penalizes larger errors more severely, RMSE is especially useful when large deviations are particularly undesirable.
Spatial Analysis	Spatial analysis examines data according to their geographic locations. In this study, it is used to detect dengue hotspots and visualize risk areas through maps and heatmaps.
Weighted Average Ensemble	A Weighted Average Ensemble is a machine learning method that combines the predictions from two or more models to create a single, optimized final prediction. This technique operates by assigning a specific weight (or importance) to each individual model's output.

XGBoost	XGBoost (Extreme Gradient Boosting) is an efficient and scalable machine learning algorithm used for supervised learning tasks, including regression and classification.

CHAPTER I

INTRODUCTION

BACKGROUND OF THE STUDY

Dengue is a quickly spreading disease with no specific treatment (Ramu et al., 2020). It is a widely transmitted infection that affects around 390 million people worldwide each year and is a major public health concern in Sri Lanka (Nagar et al., 2020). Dengue fever affects more than 120 nations worldwide. In 2019 alone, there were 5.2 million dengue cases. Although dengue is a threat to the world's population, Asia is responsible for 70% of the disease's burden around the world (W. N. Afrina Wan Muhammad Azan et al., 2024). Tropical and subtropical countries are particularly prone to dengue fever outbreaks. Dengue fever is a mosquito-borne disease that affects around 40% of the world's population.

An estimated 400 million people are infected with the dengue virus each year, resulting in 21,000 documented deaths (Nag et al., 2024). Phengsuwan et al., (2024) said that Thailand has seen an increase in dengue fever outbreaks, with over 50,000 cases reported annually in the last six decades. Until mid-December 2014, there were 71,668 documented dengue cases in 34 Indonesian provinces, with 641 of them dying (Sasongko et al., 2017).

Dengue outbreaks in the Philippines have been a persistent public health concern, with epidemics occurring every 3-4 years (Gardner & Sarkar, 2015). Genetic studies have shown that dengue virus dispersal occurs among the three main island groups of the Philippines, with Luzon identified as a major source of emergence (Sy et al., 2023). The Philippines has battled dengue for decades, with the first recorded Southeast Asian epidemic occurring in Manila

in 1954 (Ong et al., 2022). They were recorded as the highest number of dengue cases ever, in which 437,563 cases were recorded in the Philippines. Based on the data from the Department of Health (DOH), the NCR (National Capital Region) registered 6,079 dengue cases between January 1 and July 9, 2022, of which 276 were documented in Pasig with an attack rate of 3.33 (R. J. C. Cruz et al., 2024).

The female mosquitoes Aedes aegypti and Aedes albopictus are the primary carriers of the dengue virus to humans (Saikia et al., 2016). Dengue is caused by one of the four serotypes of dengue virus, the DENV-1 to DENV-4 (Barman et al., 2020). The virus causes flu-like symptoms and sometimes progresses into a life-threatening disease known as severe or hemorrhagic dengue (Spatafora et al., 2019). It can be detected by its symptoms, which include skin rashes, fever, headaches, and nausea (Minhas et al., 2019). Given its widespread prevalence, recurring outbreaks, and serious health impacts especially in tropical countries, dengue remains a major public health challenge. These conditions highlight the urgent need for forecasting to anticipate dengue outbreaks and support timely interventions.

Disease forecasting methods are important in healthcare resource allocation and preparedness for high-risk populations. Advanced data analytics and predictive modeling techniques enhance resource optimization, disease prevalence forecasting, and identification of high-risk groups (Nwoke, 2024). These approaches can help project the cumulative case burden and subsequent healthcare resource demands, particularly in rural areas that may face higher disease burdens (Miller et al., 2020). Integrating forecasts into public health decision-making requires officials to understand forecast creation,

interpretation, and past performance (Lutz et al., 2019). Studies have demonstrated the impact of forecasting models in predicting upper respiratory tract infection burdens (Lim et al., 2023) and COVID-19 hospitalizations (Castro et al., 2021).

These models enable proactive healthcare planning by estimating resource needs, such as hospital beds, equipment, and workforce requirements (Semenova et al., 2020). Predictive models can identify vulnerable populations and regions at higher risk of severe illness, allowing for targeted resource allocation (Fusillo, 2020). Recent studies have compared various forecasting methods, with machine learning approaches often outperforming traditional statistical models across different diseases and locations (Dixon et al., 2022).

Forecasting disease outbreaks helps health officials make better plans and respond faster to future outbreaks. When it comes to dengue, early warning forecasting models have shown promising results in predicting outbreaks and identifying high-risk areas in tropical regions. These models utilize climate data, including temperature, precipitation, and humidity, along with non-climate factors such as elevation and population density. In Colombia, an Early Warning Signal model successfully detected 75% of outbreaks 1-5 months ahead of time (Lee et al., 2017).

Historical case data remains the most significant contributor to predictions, accounting for about 73.63% of model performance (Long et al., 2025). A prototype dengue early warning system for Brazil demonstrated improved accuracy compared to a null model, with a hit rate of 57% for predicting high dengue risk (Lowe et al., 2016). Similarly, the GeoSeeq Dengue model forecasts outbreaks in Brazilian municipalities up to 6 months in advance

(Danko et al., 2024). These successes underscore the growing potential of dengue forecasting models as essential tools for early intervention and strategic planning, especially in countries heavily burdened by the disease.

By merging the predictions of two or more base models, ensemble learning techniques have produced excellent results in a variety of machine learning applications (Mienye et al., 2022). Vicente García et al (2019) demonstrated that ensembles with k-nn outperformed single k-nn classifiers in terms of RMSE and MAE on 129,890 airline customer samples. Amrutha K et al (2023) found that combining Random Forest with Bagging Regressor achieved lower errors and higher R^2 scores than individual regression models on wine quality prediction. Arti Pawar et al (2023) found ensemble models achieved higher accuracy and ROC-AUC scores than single models for cardiovascular disease prediction on the Cleveland Heart dataset. Ensemble models achieving 99% accuracy vs 93% for the best single algorithm (Random Forest) in sickle cell disease prediction Oluwafisayo Babatope Ayoade et al (2023). Ensemble methods also reduced prediction errors by 24-30% for infectious diseases Asmita Mahajan et al (2022).

Color-coded heatmaps are valuable tools in disease forecasting and visualization. EpiCaster, a web application, utilizes time-varying heatmaps to assess and forecast global epidemics, allowing users to visualize disease dynamics and study intervention strategies (Deodhar et al., 2015). In Kazakhstan, disease mapping techniques have been used to create risk maps, visualizing geographical distribution of diseases and forecasting risks using the Holt method (Myrzakerimova et al., 2020). A framework integrating time series analysis and vision-based methodologies for dengue spread prediction utilizes

heatmaps to visualize disease risk areas (Sharma et al., 2023). In healthcare, PHQ-9 heatmaps help clinicians efficiently understand patients' depression status and identify influencing factors (Rosas et al., 2024). Overall, heatmaps make it easier to understand and track health-related information.

Sarma et al., (2020) used the patient’s personal data, clinical data, and diagnostic data to predict. However, they said that in the future they will seek to add attributes and risk factors for dengue fever which are to be included in the dataset. Besides, climate and location-based data are also added for effective rules generation and classification. This study used three types of data including climate data such as rainfall, temperature, wind speed, etc.

OBJECTIVES OF THE STUDY

The primary objective of this study is to develop a weighted average ensemble model using Random Forest and XGBoost, combined with spatial analysis and multi-source data fusion, to visualize dengue outbreak risks in Luzon, Philippines, through a color-coded heatmap.

Specifically, the objectives of this research are the following:

1. To spatially analyze dengue outbreak patterns by integrating multi-source data comprising Epidemiological, Demographic, and Meteorological datasets using PostgreSQL/PostGIS, and by applying spatial clustering algorithms such as Getis-Ord Gi* and DBSCAN to identify statistically significant hotspot areas.
2. To develop weighted average ensemble that integrates Random Forest and XGBoost using the combined epidemiological, meteorological, and demographic datasets, along with spatial analysis results.

3. To evaluate the model's predictive accuracy using regression metrics (MAE, RMSE, R^2) and to visualize the forecasted dengue outbreak risks through a color-coded heatmap, to support geographically targeted public health interventions.

SIGNIFICANCE OF THE STUDY

The threat of dengue poses serious challenges to public health and local government units, especially in identifying and mitigating potential outbreak zones. This study aims to develop a predictive model that can forecast dengue outbreaks using an ensemble of Random Forest and XGBoost algorithms, guided by spatial data analytics. By integrating multiple datasets which are the demographic (population number and population density), epidemiological (recorded dengue cases), and meteorological data (weather related data), the study seeks to provide a data-driven approach to disease surveillance.

The significance of this research lies in its potential to support early intervention strategies, improve resource allocation, and enhance decision-making for dengue prevention and control at the community level. This study benefited many people and groups who were involved in dengue prevention and public health. Local health officers and public health agencies can use the model to identify areas that are at high risk of dengue outbreaks, helping them respond faster and more effectively. With accurate predictions, they can better plan mosquito control efforts, health campaigns, and the distribution of medical supplies.

Local government units (LGUs) can also use the information from the model to support community programs and inform residents about possible risks in their area. Researchers and students working on health, environment,

or data science can learn from the methods used in this study, especially in combining different types of data with machine learning to make useful predictions. Lastly, the general public, especially those living in dengue-prone areas, will benefit the most, as the model can help prevent the spread of the disease, reduce the number of cases, and keep communities safer and more informed.

This study supports several Sustainable Development Goals (SDGs) through its aim to predict dengue-prone areas using Weighted Average Ensemble using Random Forest and XGBoost as base model and spatial data analytics. First, it contributes to SDG 3 Good Health and Well-being by helping reduce the spread of dengue through early detection and targeted intervention, ultimately improving public health outcomes. It also aligns with SDG 9 Industry, Innovation, and Infrastructure by applying innovative technologies such as machine learning and integrating multiple types of spatial and health-related data, demonstrating how digital tools can enhance disease surveillance systems.

The study contributes to SDG 11 Sustainable Cities and Communities by identifying vulnerable locations, allowing local governments to build safer, more resilient communities with effective health preparedness strategies. Lastly, it contributes to SDG 13 Climate Action by factoring in meteorological data to understand how climate conditions influence dengue outbreaks, which supports climate adaptation efforts in public health planning. In conclusion, this study helps address the growing problem of dengue by using data and technology to predict high-risk areas. It shows how combining different types of data with machine learning can help communities stay safe and prepared. The

results of this study can be used not only by health workers and local leaders, but also by other researchers and students. Overall, this research offers a useful and practical way to support early action against dengue and protect people's health.

SCOPE AND LIMITATIONS OF THE STUDY

This study focuses on the application of spatial data analytics and machine learning techniques for the prediction of dengue outbreaks. The core objective is to develop a predictive model capable of identifying possible dengue outbreaks, enabling more informed decision-making in public health planning and response. To construct this model, the study utilizes three main types of data which are the epidemiological data, consisting of recorded dengue case counts over a period of five years (from January 2016 to December 2020), meteorological data including key weather variables such as rainfall, temperature, and humidity, and demographic data focusing on population-related factors.

The datasets serve as the foundational input for building the model. Spatial analysis will be conducted on all of the data using PostgreSQL with PostGIS, a robust tool for managing geographic and location-based information. This enables spatial querying and mapping functionalities, allowing the study to analyze how dengue patterns change over time. It also supports geovisualization, helping to reveal clustering of dengue cases and underlying environmental correlations. The study uses weighted average ensemble of Random Forest (RF) and XGBOOST machine learning algorithms. To determine when the disease is likely to occur, the study applies clustering and hotspot analysis using DBSCAN (Density-Based Spatial Clustering of

Applications with Noise) and the Getis-Ord Gi*. DBSCAN detects clusters of dengue cases based on spatial density, while Getis-Ord Gi* highlights statistically significant hotspots. The results are added to the dataset as new spatial features, such as cluster IDs and hotspot scores, which help improve the model's ability to capture spatial patterns and enhance outbreak prediction. The final results of the study are a color-coded heatmap that visualizes what months are at risk of dengue outbreaks. This visualization serves as an accessible representation of risk levels and can assist various stakeholders such as public health practitioners, policy makers, and researchers in prioritizing interventions and resource allocation.

While this study offers valuable insights and contributions to the field of disease outbreak prediction, it is important to recognize its inherent limitations, which can affect the accuracy and applicability of the results. One of the primary limitations is the quality and completeness of the data used. The accuracy of the prediction model is highly dependent on the input data. Incomplete records, inconsistencies, or missing values in epidemiological or weather data can result in biased or unreliable model outcomes. Another limitation is the lack of real-time data integration. The forecasting model is based on the data collected over a defined period. As such, it does not reflect current, real-time developments in dengue spread. This restricts the model's use for real-time alerts, which are critical in managing fast-changing disease environments. Also, some factors that affect dengue outbreaks, like mosquito population or people's habits, are not included in the current model.

In summary, this study aims to develop an ensemble of Random Forest and XGBOOST algorithms model for predicting potential dengue outbreaks with

spatial data analysis by addressing gaps encountered during the research process. It uses data on past dengue cases, weather parameters, and population, and will be used for spatial analysis and the output of it will be also used for model development. With the help of ensemble Random Forest and XGBOOST and clustering methods, the study tries to find out what months are more likely to experience dengue in the future.

The final output includes color coded heatmaps that highlight high-risk months to help in planning and response. However, the study also has some limits. The accuracy of the model depends on the quality of the data. If the data is incomplete or incorrect, the results might not be reliable. Also, the model does not work in real-time, which means it cannot give live updates. It only uses past data to make predictions. Even with these limitations, the study still gives useful information that can help in controlling dengue. These findings can be improved in the future by adding more data, using real-time updates. This way, the model can become more helpful and accurate in the long run.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

REVIEW OF RELATED LITERATURE, STUDIES, AND SYSTEMS

In this chapter, a comprehensive review of relevant literature and studies related to Spatial Analysis and Random Forest for dengue prediction and control is presented. It explores the impact of environmental and social factors, and how Random Forest algorithm have been applied to forecasting model. The chapter also examines existing systems and methodologies, highlighting their strengths and limitations. This review provides a foundation for identifying research gaps and supporting the development of an effective, data-driven approach for dengue prevention and control.

Significance of Dengue Forecasting

Predicting dengue outbreaks is crucial for effective public health interventions. Numerous studies have examined different methods to forecast outbreaks. In French Guiana, climate indicators such as sea surface temperatures and sea-level pressure were able to correctly predict 80% of outbreaks (Adde et al., 2016). In Cambodia, a forecasting tool using national surveillance data was able to explain 90% of the variance in outbreak magnitude by April, helping hospitals prepare in advance (Cousien et al., 2019). A comparative study in Indonesia found that combining meteorological data with long-term disease surveillance, including lag patterns spanning several years, was most effective for predicting dengue incidence and outbreaks (Ramadona et al., 2016).

Predictive models of dengue transmission significantly enhance early warning systems and community preparedness in high-risk areas. The Early

Warning and Response System (EWARS) uses outbreak and alarm indicators to forecast dengue outbreaks at the district level, enabling timely and efficient local responses (Hussain-Alkhateeb et al., 2018). Machine learning models, such as LASSO, have demonstrated strong performance, provided accurate three-month forecasts of dengue incidence and outperformed traditional statistical methods (Shi et al., 2016).

These models often incorporate a wide range of data sources, including recent case data, meteorological information, vector surveillance, and population statistics. Other predictors include human behavior, the physical environment, and climate factors (Satoto et al., 2019). The DengueTools research consortium has contributed significantly to the advancement of diagnostics, surveillance, prevention, and predictive modeling, offering valuable insights for public health strategies such as vaccination programs (Wilder-Smith et al., 2018).

Accurate dengue forecasting in high-risk tropical regions offers substantial public health benefits. Early warning systems can detect potential outbreaks in advance, enabling timely interventions. Identifying high-risk areas allows for more targeted prevention efforts and better allocation of limited healthcare resources (Lee et al., 2017). Forecasts assist public health officials in prioritizing resources, directing vector control efforts, and planning clinical care for severe cases. They also support improved risk communication and vaccine deployment strategies (Danko et al., 2024; Lauer et al., 2018). Seasonal forecasting systems that integrate Earth observation data, climate variables, and lagged dengue case data can predict outbreaks up to six months

in advance, enhancing preparedness and decision-making (Colón-González et al., 2021).

These tools empower authorities to take proactive measures by allocating resources and implementing prevention strategies before outbreaks occur (Riley et al., 2020; Mazhar et al., 2024). Being able to anticipate dengue outbreaks enables health workers to use funding and manpower more efficiently, helping to reduce cases and protect public health.

Spatial Analysis

The combination of spatial analysis with machine learning techniques improves the efficiency and accuracy of predictive models across various domains, including public health (Lu & Li, 2024). At the borough level in New York City, including network models generally improves influenza forecast accuracy due to highly synchronous epidemics (Yang et al., 2016). Spatial epidemiology is important for understanding disease transmission, with recent advancements in computing and statistical approaches enabling better visualization and analysis of spatial data.

Common spatial-temporal approaches include visualization, overall clustering, hot spot detection, and risk factor identification, which can improve surveillance systems, generate hypotheses, and establish prevention strategies (Lin & Wen, 2022). Spatial data analytics have a great impact in enhancing dengue fever outbreak prediction in urban environments by incorporating various factors. Machine learning models using a combination of meteorological, clinical, socioeconomic, and spatial dependence data have shown improved accuracy in forecasting dengue cases at the neighborhood level (Jain et al., 2019). Geospatial insights from multiple data sources,

including satellite imagery, public websites, and official statistics, contribute to more precise predictions at various spatial and temporal scales (Li & Dong, 2022). These advanced forecasting methods enable the creation of dynamic risk maps and spatially resolved predictions suitable for guiding targeted vector control efforts in highly urban environments (Chen et al., 2018).

Geographic Information Systems (GIS) is essential in dengue forecasting and surveillance. They are used to map spatial distribution patterns, identify high-risk areas, and develop risk models based on reported dengue cases (Purnama et al., 2022; Sulistyawati & Fatmawati, 2020). It facilitates the assessment of various risk factors, including socioeconomic and environmental variables, through spatial analysis techniques like Geographically Weighted Regression (Sulistyawati, 2020).

Recent studies highlight the significance of spatiotemporal analysis in dengue forecasting. Li et al. (2022) proposed a spatiotemporal component fusion model that incorporates mosquito abundance and spatiotemporal lags, improving prediction accuracy through effective feature representations. Li et al. (2019) integrated mosquito surveillance data with climate-driven models, showing that local vector dynamics, predictable from climate conditions, can explain dengue spatiotemporal patterns.

Filippis et al. (2016) demonstrated the integration of PostgreSQL and PostGIS in enhancing weather forecasting and spatial data management. These technologies were utilized to develop web-based tools for crop risk monitoring in Sub-Saharan Africa by processing diverse meteorological datasets to identify crop risks and support early warning systems. PostgreSQL/PostGIS geospatial databases, when deployed on Kubernetes

clusters, showed improved performance in executing computationally intensive geospatial queries compared to non-clustered environments (Sharma et al., 2021). Spatial clustering techniques have also proven effective in identifying high-risk areas for dengue transmission by incorporating factors such as social vulnerability, thereby supporting targeted control measures and urban planning (Lopes et al., 2023).

DBSCAN clustering has been successfully applied to enhance spatial load forecasting accuracy, particularly when combined with NAR neural networks (Han et al., 2020). Additionally, Zhang et al. (2024) combined the Getis-Ord Gi* statistics with DBSCAN to identify urban traffic accident hotspots, improving accuracy through parameter optimization. In the context of sales forecasting, Sozib (2024) employed DBSCAN clustering alongside ensemble modeling, achieving better predictive performance compared to traditional regression approaches.

Multi-Source Data Fusion

Researchers have developed models that integrate various data sources, including climate variables, epidemiological records, and remote sensing information (Sharma et al., 2023). These integrated approaches enable more accurate forecasting of dengue outbreaks and risk assessments. Web applications such as PICTUREE Aedes combine global risk evaluation with incidence forecasting by utilizing techniques like ensemble Kalman filters and neural networks (Yi et al., 2023). Integrated surveillance systems such as the one implemented in Foz do Iguaçu, Brazil demonstrate the practical application of these methods by incorporating both epidemiological and entomological data to support public health decision-making (Leandro & Maciel-de-Freitas, 2024).

Machine learning models have also shown promising results in predicting dengue outbreaks using historical case data from specific geographic regions.

Various studies have applied algorithms such as Support Vector Regression (SVR), Random Forest, and Naive Bayes to forecast dengue cases and related fatalities (Dey et al., 2022; Saeed et al., 2024; Thirugnanam & Hussain, 2023). Approaches incorporating climatic variables have demonstrated high accuracy in forecasting dengue outbreaks. For instance, a study in Singapore identified time and cloud cover as key predictors (Tian et al., 2024). In the Philippines, Cruz et al. (2024) found that rainfall and diurnal temperature range significantly contributed to variations in both mosquito indices and dengue cases.

Recent research has also emphasized the importance of demographic and environmental factors in predicting dengue outbreaks. Population density has been shown to significantly influence dengue incidence (Mela Firdaust et al., 2023; Sajib et al., 2024). Urbanization and mean temperature are positively associated with dengue cases, while maximum temperature and precipitation have been linked to negative effects. Socioeconomic factors, such as educational attainment and housing quality, also play a role in shaping dengue risk (Costas et al., 2023).

Weighted Average Ensemble of Random Forest and XGBOOST

Predictive accuracy, robustness, and generalizability are known to be improved by ensemble techniques, which integrate the strengths of different models (Rane et al., 2024). Ensemble learning has shown to be efficient and functional in a wide area of problem domain and substantial world application (Rincy et al., 2020). In numerous real-world applications and across a wide

range of problem domains, ensemble learning has proven to be efficient and useful (M. Shah et al., 2023). Practical applications demonstrate high accuracy rates, with some achieving up to 99.36% accuracy (Hasan et al., 2022). Therefore, ensemble learning has become a very good method for building strong machine learning systems.

C. Pop et al., (2021) found that a Weighted Average Ensemble Method provides more accurate results than individual algorithms alone when forecasting energy consumption. Galicia et al., 2019 further supported this, showing that ensemble models outperform individual members, with a dynamic ensemble achieving a remarkable Mean Relative Error of just 2% in electricity consumption prediction. Surya Narayanan K et al., (2024) showed that such an ensemble can achieve accuracy beyond 98% by combining XGBoost's data handling capabilities with Random Forest's reduced overfitting.

Financial risk assessment studies showed improved forecasting performance through weighted model averaging (Jha et al., 2021). Climate prediction research similarly validated ensemble approaches, with hybrid models producing impressive results across multiple performance metrics (Patil et al., 2024). In heart disease prediction, a weighted average ensemble achieved 93% testing accuracy (Neloy et al., 2021). Diabetes prediction studies similarly revealed ensemble models outperforming individual algorithms, with an 85% accuracy rate (Akula et al., 2019). B. Sharanya et al., (2024) achieved 92% accuracy using an ensemble of Random Forest and XGBoost with hyperparameter tuning.

One study combining traditional and mental health factors achieved 95.70% accuracy using an ensemble containing both methods (Dorraki et al.,

2022). Ariyanta et al., (2024) found their ensemble model achieved ROC-AUC of 0.9134 compared to individual Random Forest (0.9117) and XGBoost (0.9130) - representing marginal but consistent improvement. Zaman et al., (2021) reported their ensemble achieved 99.98% accuracy for heart failure survival prediction.

Ahmed et al., (2025) reported that while Random Forest individually achieved strong performance ($R^2 = 0.9789$), their ensemble approach provided enhanced stability and generalization. Reham Abdallah et al., (2024) used an ensemble for infectious disease prediction (including dengue), achieving 96.80% accuracy. Nigar et al., (2024) found that RF-XGBoost ensembles and voting regressors exhibited the best performance with lowest RMSE and MAE values in electricity load forecasting.

Color-Coded Heatmap

Color-coded heatmaps have emerged as effective tools for communicating disease prevalence and prioritizing interventions, offering advantages over standard statistical reporting. Studies have shown that heatmaps enhance interest in vaccination, increase perceived risk, and improve knowledge about infectious diseases (Fagerlin et al., 2017). They are also favored by decision-makers for their simplicity and ability to clearly convey key messages (Aung et al., 2019).

Interactive disease mapping features further improve understanding by allowing users to explore data and uncover underlying uncertainties, leading to more informed decision-making (Roberts & Gough, 2016). Disease mapping plays a vital role in identifying and prioritizing areas for health interventions. Bayesian disease mapping methods, commonly used for areal data, have

become standard practice, offering effective means to visualize and interpret spatial patterns of disease outcomes (Kang et al., 2016). These visual tools assist decision-makers in understanding and comparing risks across different scenarios. However, challenges remain in evidence-based decision-making during infectious disease outbreaks, particularly due to scientific uncertainties and competing interests (Salajan et al., 2020).

In epidemiological surveillance, a common practice shown by the World Health Organization (WHO) is to set epidemic thresholds by comparing current case counts with historical data. For example, surveillance reports from countries such as Viet Nam define the alert threshold as the five-year mean plus two standard deviations. This statistical approach helps identify real outbreaks and separate them from normal seasonal changes. Because of this, the visualization of outbreak risk, such as in a heatmap, should be based on a clear statistical threshold (World Health Organization, Western Pacific Region, 2025).

Likewise, Otshudiema et al. (2025) examined outbreak thresholds and their effects on dengue transmission using historical data collected since 2018. Their findings help public health authorities detect early warning signs of dengue outbreaks and respond more quickly before the situation worsens. According to Bowman et al. (2014), dengue outbreaks are often defined as periods when the incidence rate exceeds the mean plus two standard deviations of the same month in the previous year.

Tuan et al., 2024 found that machine learning models with climate date significantly enhanced outbreak prediction by incorporating lagged climate variables. Agarwala et al., 2024 demonstrated that an XGBoost model

enhanced with demographic features achieved the lowest Mean Squared Error of 2.6108. Ismail et al., 2022 found a Random Forest model achieved 95% accuracy when incorporating epidemiological and environmental factors. While these studies are valuable, they have not used multiple data. One of the objectives of this study is to use multiple data to forecast dengue outbreak.

RELATED LITERATURE MATRIX

Recent advancements in machine learning have demonstrated the potential of ensemble models for prediction tasks across various domains. A review of related literature, as summarized in Table 1, highlights that combining multiple algorithms often yields superior accuracy and stability compared to individual models. This approach is particularly relevant for complex forecasting challenges, such as predicting disease outbreaks, where precision is critical for effective intervention and resource allocation.

The literature provides strong evidence for the efficacy of ensembles that specifically incorporate Random Forest (RF) and XGBoost. For instance, Sharanya et al. (2024) achieved 92% accuracy in cardiovascular disease prediction using a direct ensemble of these two models. Similarly, Nigar et al. (2024) concluded that an RF-XGBoost ensemble delivered the best performance for short-term load forecasting by minimizing error rates (RMSE and MAE). The rationale for this synergy is highlighted by Surya et al. (2024), who achieved over 98% accuracy by averaging the models, they note the ensemble captures XGBoost's high accuracy while balancing it with Random Forest's strength and decreased overfitting.

Critically, C. Pop et al. (2021) found that a Weighted Average Ensemble of RF and Gradient Boosting provided more accurate predictions than when

each algorithm was used on its own, directly supporting the specific technique used in this study. The consistent success of these ensemble methods is validated across diverse fields. In finance, Ahmed et al. (2025) found that an ensemble approach provided enhanced stability and generalization for stock price prediction.

In healthcare, ensembles containing RF and XGBoost have repeatedly shown top-tier performance for predicting heart failure (Ariyanta et al., 2024) and cardiovascular disease (Dorrakil et al., 2022). For infectious disease prediction, Abdallah et al. (2024) used an ensemble model to achieve 96.80% accuracy. This body of research strongly supports the hypothesis that a weighted average ensemble of Random Forest and XGBoost is a highly promising approach for developing an accurate and reliable forecasting system.

Table 1. Matrix of Reviewed Related Literature

AUTHOR/S	TITLE	YEAR	ALGORITHM/TECHNIQUE	FINDINGS
Ahmed et al.	AI for Financial Market Prediction: Machine Learning Models for Stock Price Forecasting	2025	Random Forest Regressor (RFR), XGBoost, Support Vector Machine (SVM), LightGBM	Random Forest individually achieved $R^2 = 0.9789$, ensemble approach provided enhanced stability and generalization.
Sharanya et al.	An Efficient Ensemble Machine Learning Model for Cardiovascular Disease Prediction Using Digital Health Records	2024	XGBoost, LightGBM, Random Forest, Extra Trees, AdaBoost, Decision Tree, SVM, Logistic Regression, and Naive Bayes.	The ensemble of Random Forest and XGBoost achieved 92% accuracy with hyperparameter tuning.

Ariyanta et al.	Ensemble learning approaches for predicting heart failure outcomes: A comparative analysis of feedforward Neural Networks, Random Forest, and XGBoost.	2024	Feedforward Neural Network (FNN), Random Forest, and XGBoost.	Their ensemble model achieved ROC-AUC of 0.9134 compared to individual Random Forest (0.9117) and XGBoost (0.9130).
Abdallah et al.	Leveraging AHP and transfer learning in machine learning for improved prediction of infectious disease outbreaks	2024	Random Forest, XGBoost, Gradient Boosting	Their ensemble model achieved 96.80% accuracy.
Nigar et al.	A Comparative Analysis of Ensemble Methods for Short-Term Load Forecasting	2024	Ensemble of RF and ETR, RF and XGBoost, XGBoost and ETR	They found that RF-XGBoost ensembles exhibited the best performance with lowest RMSE and MAE.
Surya et al.	A Machine learning approach for crop and fertilizer using ensemble model with XGBoost and Random Forest algorithms.	2024	XGBoost and Random Forest	By averaging predictions from both algorithms, the model achieves a high level of accuracy beyond 98%, outperforming conventional single model systems. The great accuracy of XGBoost and its data handling capabilities are captured by this ensemble design, which also balances it with Random Forest's strength and decreased overfitting.
Dorrakil et al.	Cardiovascular disease risk prediction via machine learning using mental health data	2022	Decision Tree, Random Forest, XGBoost, Support Vector Machine (SVM), Deep Neural Network (DNN)	The model achieved 95.70% accuracy using an ensemble containing both algorithms.

C. Pop et al.	Forecasting the Short-Term Energy Consumption Using Random Forests and Gradient Boosting.	2021	Random Forests and Gradient Boosting algorithms	Both algorithms were used separately to predict energy consumption, and then their results were combined using a Weighted Average Ensemble Method. After comparing the results, it turned out that the ensemble method gave more accurate predictions than when each algorithm was used on its own.
Galicia et al.	Multi-step forecasting for big data time series based on ensemble learning.	2019	Decision Tree, Gradient Boosted Trees and Random Forest	They discovered that the ensemble model was the most accurate, producing a very promising outcome for large-scale time series prediction.

SYNTHESIS AND RELEVANCE OF THE STUDY

Dengue fever is prevalent in tropical and subtropical locations, where Aedes mosquitoes, the virus's primarily vectors, live in warm and humid conditions (Nag et al., 2024). It affects more than 12 million individuals in the world (Hossain & Nishu., 2024). Dengue outbreaks have multi-sectoral impacts, including increased hospital waiting times and financial strain on low-income families. The economic burden of dengue disproportionately affects low-income families and informal sector workers, posing an equity challenge (Ladner et al., 2017; Marczell et al., 2024). Climate variability plays a crucial role in dengue transmission, as warmer temperatures accelerate vector and pathogen metabolism, while erratic rainfall patterns may increase suitable breeding sites (Permatasari & Masjud, 2024).

Regarding the seriousness of dengue cases among the public several quantitative predictive approaches have been developed for forecasting outbreaks of dengue (Mustaffa et al., 2018). In tropical areas, machine learning

(ML) has demonstrated promising outcomes in forecasting dengue outbreaks. Research has shown that machine learning models, especially when utilizing surveillance data, perform better than traditional statistical techniques in predicting dengue cases and outbreaks (Benedum et al., 2020; Chen & Moraga, 2024).

The reviewed literature emphasizes that dengue continues to surge in tropical and subtropical regions where environmental conditions favor the growth of mosquitoes. Studies show that rising temperatures, persistent humidity, and irregular rainfall significantly increase mosquito breeding sites and virus transmission. Because dengue patterns vary greatly between localities, spatial analysis has become a key component in outbreak studies. Techniques such as spatial clustering and hotspot detection allow researchers to pinpoint high-risk communities and monitor how infection spreads geographically over time. By incorporating geospatial datasets such as locality, case reports and climate variables forecasting becomes more precise, enabling customized intervention strategies rather than broad, less effective control measures.

The literature further shows a growing adoption of machine learning for dengue forecasting, with Ensemble models combining Random Forest and XGBoost and other machine learning model demonstrate superior results across multiple studies. These models effectively capture nonlinear disease behavior and reduce both bias and variance by leveraging the complementary strengths of tree-based learners. The integration of performance-optimized weights improves prediction stability, especially in regions with fluctuating cases. To support practical decision-making, researchers increasingly use

color-coded heatmaps to visualize detected outbreaks and predicted risk intensities at the regional and locality levels. These visual analytics help health authorities quickly identify emerging hotspots, allocate resources efficiently, and implement timely mosquito control and community awareness campaigns. Overall, the synthesis of spatial analytics, ensemble forecasting, and outbreak-focused heatmap visualization strengthens early-warning capabilities and advances dengue control efforts in the most affected areas.

CONCEPTUAL FRAMEWORK

As shown in Figure 1, this study will develop a Weighted Average Ensemble of Random Forest and XGBoost to forecast dengue outbreaks in selected regions of Luzon, Philippines specifically Region I, Region III, Region IV-A, Region V, CAR, and NCR. The framework begins with the collection of three primary data types: epidemiological (dengue cases), demographic (population), and meteorological (weather conditions).

Epidemiological approaches are important in predicting dengue outbreaks by analyzing various factors. Recent studies highlight the importance of demographic factors in predicting dengue outbreaks. Population density is positively associated with dengue incidence (Firdaust et al., 2023; Sajib et al., 2024). Different types of weather parameter can contribute to the spread of dengue, necessitating enhanced surveillance (Almeida et al., 2025). Seasonal weather patterns, specifically meteorological factors, affect mosquito population dynamics, which is essential for designing effective vector control strategies (Baafi & Hurford., 2025).

These datasets will first be processed through spatial analysis. Once the spatial outputs are generated and merged with the epidemiological,

demographic, and meteorological datasets, the combined data will undergo preprocessing, which includes data cleaning and feature engineering to ensure high-quality inputs for model training. After preparation, the structured dataset is used to train two base models: Random Forest and XGBoost. Their prediction outputs are then integrated using a Weighted Average Ensemble approach, designed to leverage the strengths of both models and improve overall forecasting accuracy.

The model's predictive performance for future dengue cases will be evaluated using regression metrics including Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Root Mean Squared Error (RMSE). The output will be predicted dengue case counts and heatmap visualizations identifying high-risk outbreak areas across Luzon to support decision-making and proactive public health interventions.

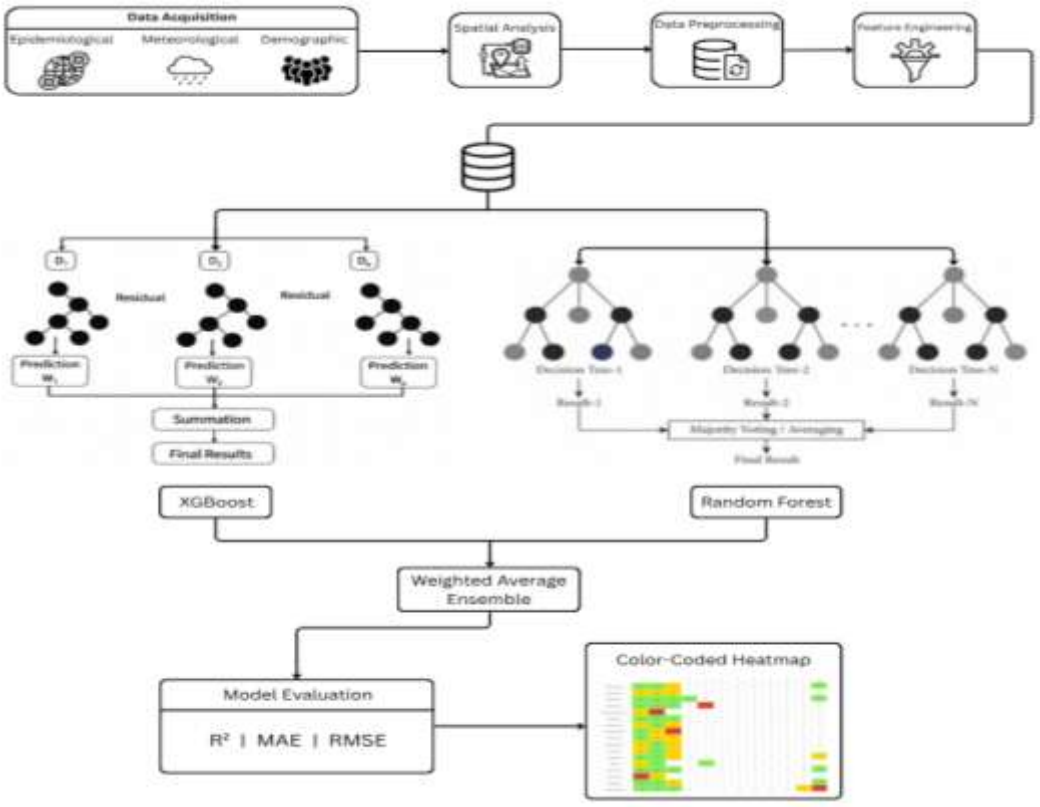


Figure 1. Conceptual Framework

CHAPTER III

METHODOLOGY

STUDY AREA

This study, conducted by Computer Science students from Laguna State Polytechnic University Los Baños Campus, focuses on some Regions of Luzon such as Region I, Region III, Region IV-A, Region V, CAR and NCR as the primary location for model development. As the largest and most diverse island in the Philippines, Luzon offers a wide range of environmental and geographic conditions that are essential for building a robust and representative predictive model.

The island includes coastal zones, lowland plains, highland regions, and both urban and rural communities, each experiencing varying temperature levels, rainfall patterns, and humidity depending on elevation and geographic position. With its tropical climate, Luzon undergoes distinct wet and dry seasons, and these climatic differences are further influenced by monsoons and localized weather systems. Such environmental variability is crucial for developing accurate models, especially in forecasting events that are sensitive to spatial and temporal changes, such as disease outbreaks. The diversity of Luzon's physical and climatic conditions ensures that the model developed from this area will be adaptable and relevant for a wide range of settings.

MATERIALS

The study used various resources and materials, including software, hardware, libraries, and tools, which was essential for the development and completion of the model.

SOFTWARE

This study utilized a range of software tools essential for data processing, model development, machine learning, and spatial analysis. These include programming environments, database systems, and statistical libraries that support the integration, analysis, modeling, and visualization of epidemiological, demographic, and meteorological data.

Visual Studio Code

A widely used, lightweight source-code editor that supports multiple programming languages and tools through extensions. It acts as the central platform for writing, editing, and managing Python code. With features like syntax highlighting, Git integration, and debugging tools, it helps streamline development and maintain organized scripts throughout the data processing and modeling stages.

Python (Jupyter Notebook)

An interactive development environment that blends live code, outputs, visualizations, and explanatory text in a single document. It is useful for conducting experiments with datasets, building and evaluating models, and documenting the workflow clearly. Its step-by-step format makes it ideal for iterating through analysis while keeping the process easy to follow and reproducible.

PostgreSQL

A powerful and open-source relational database system that efficiently manages structured datasets. It is particularly effective in handling large-scale data storage, allowing secure access, modification, and querying of data. This

tool is essential for storing raw and processed data that includes both regular tabular data and geospatial attributes.

PostGIS

An extension of PostgreSQL that enables the database to handle geographic and spatial data types. It adds functionalities for storing, querying, and performing spatial operations such as distance calculations, area measurements, and spatial joins. It plays a critical role in managing geospatial information that is later used for mapping or spatial analysis.

QGIS (Quantum Geographic Information System)

An open-source desktop GIS application used to view, analyze, and create maps from spatial datasets. It allows users to overlay multiple layers of data, apply spatial filters, and extract insights from location-based patterns. It provides a visual representation of spatial data that supports deeper interpretation and supports decision-making through map-based outputs.

Scikit-Learn

A Python library that provides simple and efficient tools for data mining and machine learning. In this study, Scikit-Learn was used for model development, training, and evaluation. Specifically, it provided functions for data splitting (train_test_split, KFold, TimeSeriesSplit), model performance assessment (mean_squared_error, mean_absolute_error, r2_score, make_scorer), and ensemble learning (RandomForestRegressor). These tools enabled systematic model optimization and performance comparison across different predictive approaches.

XGBoost

A Python library designed for gradient boosting, widely used for both regression and classification tasks. It is optimized for speed and performance through parallel processing, regularization, and advanced tree-pruning techniques. XGBoost enhances prediction accuracy by sequentially improving weak learners and minimizing overfitting, making it one of the most popular frameworks for building robust and scalable machine learning models in data analysis and predictive modeling.

Optuna

A Python framework for automatic hyperparameter optimization using Bayesian optimization and other search algorithms. It efficiently explores the hyperparameter space through a trial-based approach that balances exploration and exploitation. Optuna's key features include dynamic pruning of unpromising trials, parallel execution, and easy integration with popular machine learning libraries such as Scikit-Learn and XGBoost. It helps improve model performance by systematically identifying optimal hyperparameter combinations with minimal manual tuning.

Pandas

A high-level Python library designed for data manipulation and analysis. It offers data structures like DataFrames that simplify operations such as merging datasets, filtering records, handling missing values, and computing statistics. This tool is often the starting point for preparing data before applying machine learning models.

NumPy

A core scientific library in Python that supports large, multi-dimensional arrays and matrices, along with mathematical functions to operate on them. It provides the foundation for numerical computations used in data analysis and is frequently used in combination with Pandas and Scikit-learn to perform efficient data transformations and calculations.

Psycopg2

A PostgreSQL database adapter for Python that enables direct interaction between Python programs and PostgreSQL databases. It is used to establish database connections, execute SQL queries, and manage transactions efficiently. In this study, psycopg2 facilitates the retrieval of records with missing geographic data and updates the database with newly acquired latitude and longitude values.

Geopy

A Python library designed for geocoding and reverse geocoding tasks, converting place names or addresses into geographic coordinates. It provides access to several geocoding services, including Nominatim from OpenStreetMap. In this research, Geopy was utilized to obtain the latitude and longitude of localities in the Philippines, serving as a critical component in enhancing the spatial completeness of the dataset.

Geopy.exc (GeocoderTimedOut)

A submodule of Geopy that provides exception handling for geocoding-related errors. The GeocoderTimedOut exception ensures that the program can handle delays or service interruptions gracefully. In this study, it was applied to

manage timeout errors during geocoding requests and to retry failed queries, improving the reliability of the data collection process.

tqdm

A Python library that adds progress bars to loops and iterative processes. It provides real-time visual feedback on the progress of long-running tasks. In this study, tqdm was used to monitor the geocoding process of multiple localities, allowing better tracking of progress and ensuring transparency during data preparation. tqdm improves usability by showing the estimated time remaining and processing speed during execution. This helps identify slow or stalled steps in the geocoding workflow and supports better debugging and performance monitoring.

Time

A built-in Python library that provides functions related to time management, such as delays and time tracking. In this research, the time library was specifically used to introduce short pauses when retrying geocoding requests after timeouts, ensuring smoother execution and preventing service overload. The time library also enables measurement of execution duration for specific processing stages. This allows assessment of computational efficiency and supports controlled request pacing to comply with external service limits.

GeoPandas

A Python library that extends the capabilities of pandas to handle geospatial data. It integrates well with shapely, fiona, and other GIS libraries, allowing the storage, manipulation, and analysis of geographic objects such as points, lines, and polygons. In this study, GeoPandas was used to load spatial

data from PostgreSQL, manage geographic attributes, and prepare datasets for spatial analysis and hotspot detection.

Shapely

A Python package for the manipulation and analysis of planar geometric objects. It supports geometric operations such as intersections, unions, buffering, and coordinate handling. In this study, Shapely was utilized through its Well-Known Binary (WKB) module to interpret and handle geometry data stored in the PostgreSQL database.

SQLAlchemy

A Python SQL toolkit and Object Relational Mapper (ORM) that facilitates efficient interaction with relational databases. It provides a high-level interface for building SQL queries and managing database connections. In this research, SQLAlchemy was employed to connect Python to PostgreSQL, allowing direct retrieval of spatial and non-spatial records for analysis.

PySAL (Python Spatial Analysis Library)

A comprehensive open-source library for spatial data science. It provides tools for spatial weights construction, exploratory spatial data analysis, and spatial econometrics. In this study, PySAL was used to perform hotspot analysis by constructing spatial relationships between locations and applying the Getis-Ord Gi* statistics to detect significant clusters of dengue cases.

Principal Component Analysis (PCA)

A module in Scikit-learn for performing Principal Component Analysis (PCA). PCA reduces high-dimensional datasets into fewer uncorrelated components while preserving as much variance as possible. In this study, PCA was used to combine epidemiological, demographic, and meteorological

features into principal components, which were then analyzed to identify spatial hotspots.

HARDWARE

Table 2. Hardware Specifications for Model Development

Specifications	Details
Processor	Intel(R) Core(TM) i5-1035G1 CPU @ 1.00GHz
GPU	NVIDIA GeForce MX330
Memory	8.0 GB
Solid State Drives	512 GB / NVMe INTEL SSDPEKNW512G8
Operating System	Windows 11 Home
System Type	64-bit operating system, x64-based processor

The primary hardware utilized for this study is the system presented in Table 2, equipped with specifications capable of supporting intensive data processing, machine learning model training, and spatial data analysis. The device is equipped with sufficient memory, storage, and processing power to handle the datasets involved in integrating epidemiological, meteorological, and demographic. Furthermore, its dedicated graphics processing unit (GPU) supports hardware-accelerated computations and visualization tasks, which are essential for spatial modeling.

The device's solid-state drive (SSD) ensures faster read/write operations, which significantly improves the efficiency of data loading, model training cycles, and geospatial computations. The system runs on a 64-bit operating environment, enabling compatibility with modern software libraries and frameworks used in scientific computing and machine learning. This hardware setup ensures reliable performance and responsiveness during the entire research and development process. The system also supports parallel

processing and multitasking, allowing simultaneous execution of spatial queries, machine learning algorithms, and data preprocessing pipelines. It is capable of managing datasets without performance degradation, supports continuous testing and evaluation of models.

RESEARCH DESIGN

This study adopts a theoretical, quantitative, and predictive research design to develop a dengue outbreak forecasting model using spatial data analytics and machine learning. It emphasizes the construction and evaluation of a predictive model within a controlled environment using historical data, rather than implementing a real-time system. The study employs a weighted average ensemble that integrates Random Forest and XGBoost as base learners to estimate future dengue cases. This ensemble approach leverages Random Forest's robustness against noise and XGBoost's gradient boosting efficiency to enhance model stability, accuracy, and generalization. Strategies include Bayesian hyperparameter tuning to optimize ensemble performance and minimize overfitting.

Historical data from epidemiological, meteorological, and demographic sources will be preprocessed by handling missing values, normalizing numerical variables, creating lag variables, and merging datasets based on geographic location and date. Spatial analysis will be conducted using DBSCAN to identify groupings of high-risk areas, while the Getis-Ord Gi* statistics will detect significant dengue hotspots. These outputs will be encoded as additional features to incorporate geographic context into the model.

The weighted average ensemble will be trained on structured datasets combining temporal, environmental, demographic, and spatial variables. Both

Random Forest and XGBoost models will serve as base models, and their predictions will be combined using optimal weights. These weights will be determined through Optuna hyperparameter optimization to minimize prediction error and produce the final dengue case forecasts. Evaluation metrics will include MAE, RMSE, and R^2 to assess accuracy.

Color-coded heatmaps will visualize the predicted dengue cases from January to December 2020. Each month's prediction will be compared against the outbreak threshold determined using the World Health Organization (WHO) formula to identify potential outbreak periods. The color scheme will indicate prediction accuracy: green for high accuracy, yellow for moderate accuracy, and red for low accuracy. This visualization supports clear temporal analysis of outbreak patterns and helps assess model reliability across different time periods.

DATA COLLECTION ACQUIRE

This study utilized data from multiple official sources covering January 2016 to December 2020 for the development of a Weighted Average Ensemble model to forecast dengue cases in selected regions of Luzon. Epidemiological data on reported dengue cases were obtained from the Humanitarian Data Exchange (HDX) via the Department of Health (DOH), an open-source platform providing accessible public health datasets. Meteorological data, including temperature, rainfall, humidity, wind speed, and wind direction, were collected from the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) through a formal request process.

Demographic data such as population size and density were requested from the Philippine Statistics Authority (PSA) via email to provide context

regarding area-specific vulnerability and risk. All datasets were either publicly available or obtained through official communication with the relevant agencies, ensuring the reliability and appropriateness of the information used in the study.

Table 3. Epidemiological Data from Humanitarian Data Exchange (DOH)

Column	Sample Size
Cases	11700

Table 3 contains 11,700 recorded dengue cases corresponding to the period from January 2016 to December 2020. Each entry represents the number of confirmed cases in specific localities and time periods, providing a direct measure of disease occurrence. This dataset serves as the primary indicator of dengue outbreak intensity and is essential for analyzing temporal patterns over the five-year period.

Table 4. Demographic Data from Philippine Statistics Authority (PSA)

Column	Sample Size
Population	11700
population_density	11700

Table 4 consists of 11,700 records aligned with Table 3, including total population counts and population density for each locality. These demographic variables provide context for understanding disease transmission, as areas with higher population or density may have increased exposure to dengue. Integrating this data with Table 3 supports risk assessment and comparative analysis across localities over time.

Table 5. Meteorological Data from PAGASA

Column	Sample Size
rainfall	11700
relative_humidity	11700
temp_max	11700
temp_min	11700
wind_speed	11700
wind_direction	11700

Table 5 contains 11,700 entries corresponding to the same region and localities and periods as Tables 3 and 4. It includes key environmental variables such as rainfall, relative humidity, maximum and minimum temperatures, wind speed, and wind direction. These factors influence mosquito breeding and virus transmission. Combining this dataset with Tables 3 and 4 enables a comprehensive analysis of the epidemiological, demographic, and environmental drivers of dengue outbreaks over the five-year period.

MODEL DEVELOPMENT

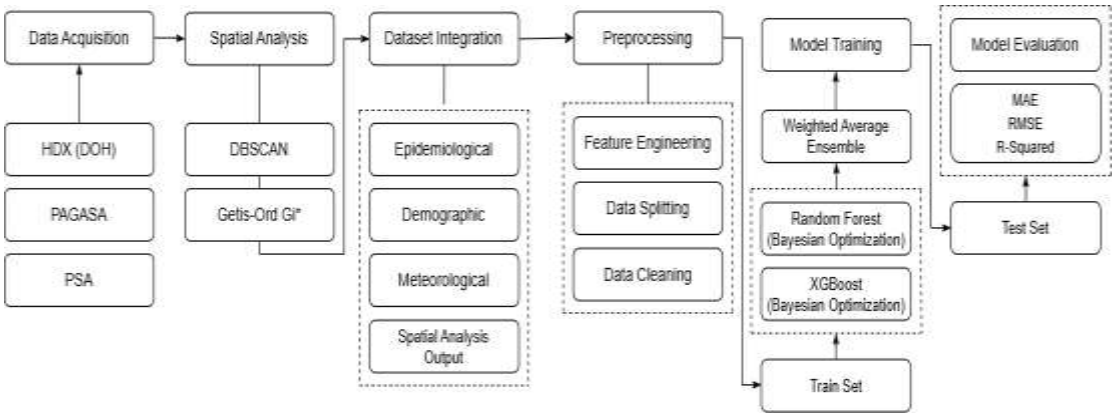


Figure 2. Model Development Diagram

Data Acquisition

In this study, the data acquisition process drew from three main sources: epidemiological records (Humanitarian Data Exchange - DOH), demographic

information (Philippine Statistics Authority), and meteorological data (PAGASA). Integrating these sources allowed the construction of a comprehensive dataset that reflects disease patterns, population dynamics, and environmental conditions relevant to dengue transmission. This multi-source integration ensured that the Ensemble of Random Forest and XGBoost was trained on diverse and complementary factors, strengthening its capacity to generate reliable forecasts.

Epidemiological Data

Epidemiological data were obtained from the Humanitarian Data Exchange, which contained records of reported dengue cases across various regions and localities in the Philippines. The dataset included time-based attributes such as dates, as well as spatial identifiers like region and locality, allowing for temporal and spatial tracking of disease incidence. These records served as the primary outcome variable for the study, providing the baseline information needed to link dengue case patterns with environmental and demographic factors.

Demographic Data

Demographic information was sourced from the Philippine Statistics Authority (PSA). The dataset included key variables such as population and population density, which were incorporated into the analysis to reflect human exposure and transmission risk factors. By integrating demographic characteristics with epidemiological data, the study accounted for variations in community size and settlement patterns, which influence the likelihood and intensity of dengue outbreaks.

Meteorological Data

Meteorological data were acquired from the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA). The dataset consisted of climate-related variables, including rainfall, relative humidity, minimum and maximum temperature, wind direction, and wind speed. These variables were selected because weather conditions directly affect mosquito breeding, survival, and biting activity, making them critical predictors in dengue forecasting. Incorporating meteorological data allowed the study to capture environmental drivers of disease transmission within the predictive modeling framework.

While many studies forecast dengue using either meteorological data or epidemiological case histories, this research presents a more holistic model. The novelty lies in the multi-source data fusion of three distinct domains, epidemiological (dengue cases), demographic (population and population density), and meteorological (weather related data). This integration allows the model to capture real-world interactions, such as how weather might uniquely impact densely populated areas to influence future outbreaks.

Spatial Analysis

This is the process of examining data with a geographic or spatial component to identify patterns, relationships, and trends. It uses specific techniques to study objects, events, or phenomena based on their location, attributes, and relationships with other features. It is a broad field that includes methods like spatial statistics and clustering. A key technique within spatial analysis is the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. DBSCAN is an unsupervised clustering algorithm

specifically designed to discover clusters of varying shapes and sizes in a dataset by grouping together points that are closely packed. A significant advantage of DBSCAN is its ability to identify and mark as outliers those points that are situated in low-density regions, effectively isolating them from the main clusters. The algorithm operates by first identifying "core" points, defined as those having a minimum number of other points (minPts) within a specified distance (epsilon). From these core points, it then expands clusters, adding all reachable points to the same cluster. This process is particularly useful for discovering patterns of point density within a dataset, addressing the question of where dense groups of points are located.

Complementing clustering methods, the Getis-Ord Gi* statistic is a crucial spatial statistics tool used for hot spot analysis. This method calculates a z-score and a p-value for each feature in a dataset to determine if high-value or low-value clusters are statistically significant. The calculation for each feature takes into account not only its own value but also the values of all its neighboring features. The resulting z-score provides a measure of the intensity of clustering. The primary purpose of the Getis-Ord Gi* statistic is to identify statistically significant spatial clusters of high values (hotspots) and low values (coldspots), thereby answering the question of whether high-value or low-value features are clustered more than what would be expected by random chance.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Table 6 shows the results of clustering using different eps values. To identify patterns and high-risk areas of dengue cases, spatial analysis was performed using Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and hotspot detection through the Getis-Ord Gi* statistics. The

latitude and longitude of all localities in the dataset were obtained using a Python script with the geopy library to automatically retrieve geographic coordinates. This process produced a complete list of localities with their corresponding spatial points. QGIS was then used to run the DBSCAN algorithm and group nearby cases together.

Since the data were at the provincial and city levels, the eps (distance) parameter was set to 50 km (0.450000 degrees) so that only points within this range could form clusters. For the minPts parameter, the rule of thumb of $d + 1$ was followed, where d is the number of dimensions; in this case, $d = 2$, resulting in a value of 3. However, values of 3 and 4 produced too many small and insignificant clusters, while values of 6 or higher detected clusters only in areas with very high case counts. To achieve a balanced result, minPts was set to 5, which produced clusters that were meaningful without being overly restrictive. DBSCAN then generated the LOCALITY_CLUSTER and CLUSTER_SIZE values for each locality.

Table 6. Clustering Results for Different eps Values

eps (km)	Number of Clusters	Min Cluster Size	Max Cluster Size
10	32	260	4,420
20	28	260	4,680
30	24	260	4,940
40	22	260	5,200
50	15	260	7,540

Getis Ord Gi*

After clustering, hotspot analysis was performed using the Getis-Ord Gi* statistics in Python. Prior to the analysis, all features in the dataset were standardized using Principal Component Analysis (PCA) to ensure fair

comparison. The k-nearest neighbors (kNN) method with $k = 5$ was applied to ensure that each locality had at least five neighbors. A locality was classified as a hotspot if its z-score was +1.645 or higher and as a cold spot if its z-score was −1.645 or lower. A p-value threshold of 0.10 was used to determine statistical significance. Finally, the results were used to generate a heatmap highlighting areas with high and low dengue case concentrations.

Dataset Integration

The datasets were merged to support the analysis of dengue patterns. Epidemiological, demographic, and meteorological data were linked through common geographic identifiers such as region and locality, along with temporal markers (Year, Month, Day). Spatial analysis results were also incorporated to reflect clustering and hotspot areas, strengthening the geographic perspective of the dataset. This integration aligned health outcomes with environmental, demographic, and spatial conditions, providing a solid basis for identifying trends and training the model.

Data Cleaning

The data cleaning process was conducted to ensure the accuracy, reliability, and consistency of the dataset prior to analysis and model development. Initially, the year, month, and day columns were combined to form a unified date variable, which enabled proper chronological ordering of records by region and locality. Missing data in critical variables such as dengue cases, population, population density, rainfall, relative humidity, temperature, wind speed, wind direction, and spatial attributes were systematically identified and removed to prevent analytical distortion.

For remaining numeric attributes containing missing values, the K-Nearest Neighbors (KNN) Imputer was employed with five neighbors to estimate values based on the most similar data points. This method maintained the local statistical structure of the data and reduced potential biases introduced by arbitrary imputation. Furthermore, outlier handling was applied through an enhanced interquartile range (IQR) approach using varying sensitivity levels—strict, moderate, and lenient—depending on the variable type. This ensured that extreme values were effectively controlled without discarding valid data points.

Table 7. Descriptive Statistics

Column	N	Missing (%)	Mean	SD	Median	Min	Max
population	11160	0.0	1072925.98	1056347.19	629616	65227	3340254
population_density	11160	0.0	9868.96	12471.57	2816.00	28.00	42488.50
rainfall	11160	0.0	47.82	58.41	23.75	0	202.20
relative_humidity	11160	0.0	78.48	8.38	79.43	50.29	98.29
temp_max	11160	0.0	31.18	2.82	31.61	24.54	38.47
temp_min	11160	0.0	23.74	2.34	24.34	18.13	29.10
wind_speed	11160	0.0	2.04	0.92	1.86	0	5.79
wind_direction	11160	0.0	179.54	70.09	187.14	8.57	360
locality_cluster_number	11160	0.0	5.02	3.64	3.00	1.00	15
cluster_size	11160	0.0	5015.11	3401.47	7540.00	260	7540
g_score	11160	0.0	-0.12	0.24	-0.12	-0.85	0.63
p_value	11160	0.0	0.24	0.16	0.21	0	0.50

Table 7 presents the descriptive statistics of the study variables based on 11,160 observations with no missing data across all columns. Population shows substantial variability (Mean = 1,072,925.98; SD = 1,056,347.19), indicating large differences in locality sizes, which is also reflected in population density with a wide range from 28.00 to 42,488.50 persons per km². Climatic variables indicate generally moderate conditions, with average rainfall of 47.82 mm, high relative humidity (Mean = 78.48%), and mean maximum and minimum temperatures of 31.18 °C and 23.74 °C, respectively.

Wind conditions are relatively stable, with low mean wind speed (2.04 m/s) and a broadly distributed wind direction. Spatial clustering variables show an average locality cluster number of 5.02 and a mean cluster size of 5,015.11, suggesting heterogeneous spatial grouping across areas. The Getis-Ord Gi* statistics indicate weak spatial clustering overall, as reflected by the near-zero mean g_score (−0.12) and moderate mean p_value (0.24), implying limited statistically significant hotspot or coldspot patterns in the dataset.

Data Splitting

In this study, time series cross-validation was applied using the TimeSeriesSplit method with five splits to evaluate the predictive performance of the model while preserving the temporal order of the data. Unlike random k-fold cross-validation, this approach ensures that training data always precedes testing data, which is crucial for time-dependent datasets such as dengue case counts. Table 8 summarizes the distribution of training and testing indices across the five splits. For each split, the training set gradually increases in size, starting from the first 1,860 observations in Split 1 up to 9,300 observations in Split 5, while the test set consistently contains 1,860 subsequent observations.

The indices indicate the first and last positions of the samples included in each set, showing that no future information is used to predict past values, thus maintaining the integrity of time-dependent modeling. For the final evaluation, the last split was selected, where the training set consisted of observations 0 to 9,299 and the test set included observations 9,300 to 11,159. This approach allows the model to learn from a growing window of historical data and be validated on the immediately following unseen data, which reflects realistic forecasting conditions for dengue outbreak prediction.

Table 8. TimeSeriesSplit train-test distribution

Split	Train Indices	Train Samples	Test Indices	Test Samples
1	0 → 1859	1860	1860 → 3719	1860
2	0 → 3719	3720	3720 → 5579	1860
3	0 → 5579	5580	5580 → 7439	1860
4	0 → 7439	7440	7440 → 9299	1860
5	0 → 9299	9300	9300 → 11159	1860

Feature Engineering

The feature engineering phase was implemented to extract meaningful temporal, spatial, and statistical features that capture the dynamic patterns of dengue transmission across different localities. Lag features were generated at intervals of one, two, four, eight, and twelve weeks to represent delayed effects of previous dengue cases on future occurrences. These lag variables helped model the autocorrelation present in epidemiological data and provided temporal continuity for prediction. Also, rolling window statistics were computed over three-, seven-, and fourteen-week intervals to summarize short, medium,

and long-term case fluctuations. These included rolling mean, standard deviation, maximum, and minimum values of past dengue cases, ensuring representation of variability and persistence within each locality. All rolling operations were performed within region–locality groups to maintain spatial relevance and prevent leakage across different areas.

To further capture recent changes while smoothing epidemiological noise, exponential moving averages (EMA) with smoothing factors of $\alpha = 0.3$ and 0.5 were introduced. These features increased the model's ability to detect gradual shifts in dengue incidence by placing stronger weight on recent observations. Moreover, risk-related spatial indicators were derived from geostatistical outputs: a high-risk area flag was encoded based on g-score exceedance beyond the 70th percentile, and a significant cluster flag was constructed from p-values lower than 0.05 to identify statistically meaningful transmission clusters. Seasonal patterns were incorporated using sinusoidal encodings of both week and month cycles, enabling the model to capture annual periodicity in dengue outbreaks.

After all feature derivations, missing values were removed to ensure dataset consistency. Categorical attributes such as region, locality, and status were encoded using one-hot encoding to ensure model compatibility. Collectively, these engineered features enhanced the dataset's representational capacity by incorporating temporal dependencies, spatial risk patterns, and seasonal behavior. As a result, the final dataset was structured, information-rich, and suitable for developing reliable and spatially aware dengue forecasting models.

Model Training

In this study, a weighted average ensemble model combining Random Forest and XGBoost was developed to forecast dengue outbreak incidence using integrated epidemiological, demographic, and meteorological datasets, supported by spatial analysis.

Table 9. Hyperparameter search space of Random Forest

Hyperparameter	Value
n_estimators	100, 1000
max_depth	3, 20
min_samples_split	2, 20
min_samples_leaf	1, 15
max_features	sqrt, log2, None
bootstrap	True, False

Table 9 presents the hyperparameter search space defined for the Random Forest model during optimization using Bayesian search in Optuna, where the optimization process was executed for 120 trials to identify the best-performing combination. The n_estimators parameter ranges from 100 to 1000 and determines the number of decision trees, ensuring a strong balance between predictive performance and computation time. The max_depth parameter, set between 3 and 20, controls the maximum depth of each tree to limit overfitting while maintaining learning capacity.

The min_samples_split parameter, ranging from 2 to 20, specifies the smallest number of samples required to split a node, helping regulate how the model adapts to data patterns. The min_samples_leaf parameter ranges from 1 to 15 and sets the minimum number of samples in terminal nodes, enabling better generalization. The max_features parameter, with options such as sqrt,

log2, or None, influences how many features are considered when selecting splits, which helps reduce correlation between trees and enhances ensemble diversity.

Finally, the bootstrap parameter allows the decision between sampling with replacement (True) or using the entire dataset (False), which affects variability during training. Through these carefully selected hyperparameter ranges and the 120 Optuna trials, Table 9 ensures a robust optimization setup for producing a well-tuned Random Forest model that serves as a reliable base model in the weighted average ensemble.

Table 10. Hyperparameter search space of XGBoost

Hyperparameter	**Value**
n_estimators	100, 1000
max_depth	3, 20
learning_rate	0.01, 0.5
subsample	0.5, 1.0
colsample_bytree	0.5, 1.0
gamma	0, 10
reg_alpha	0, 5
reg_lambda	0, 5

Table 10 defines the hyperparameter search space used to optimize the XGBoost model through Bayesian search using Optuna, with a total of 180 trials conducted to obtain the best-performing configuration. The n_estimators parameter ranges from 100 to 1000 and controls the number of boosting rounds, directly affecting model performance and training cost. The max_depth parameter ranges from 3 to 20 and determines how deep each tree can grow, helping balance complexity and generalization. The learning_rate parameter varies between 0.01 and 0.5, regulating the contribution of each tree to the final

prediction and ensuring stable learning. The subsample parameter ranges from 0.5 to 1.0, controlling the proportion of training data sampled per boosting iteration to reduce overfitting.

The colsample_bytree parameter, also set between 0.5 and 1.0, determines the fraction of features used to construct each tree, encouraging model diversity and preventing excessive reliance on specific features. The gamma parameter ranges from 0 to 10 and sets a minimum loss reduction threshold required to make a further split, promoting more efficient and meaningful branching. Finally, the regularization parameters reg_alpha and reg_lambda, each ranging from 0 to 5, apply penalties to model weights to improve generalization and reduce overfitting. Through these hyperparameter ranges and 180 optimization trials, Table 10 ensures a robust tuning process that enhances the XGBoost model's predictive accuracy and reliability as a base model in the weighted average ensemble.

Table 11. Search Space for Base Model Weights

Base Model	Weight
Random Forest (w_rf)	0.0, 1.0
XGBoost (w_xgb)	1.0 – w_rf

Table 11 defines the search space for determining the optimal weights of the base models used in the weighted average ensemble, optimized using Optuna with 200 trials while minimizing the Root Mean Squared Error (RMSE) as the objective metric. The weight for the Random Forest model (w_rf) ranges from 0.0 to 1.0, enabling the ensemble to adjust its reliance on Random Forest based on performance. The weight of the XGBoost model is defined as 1.0 –

w_rf, ensuring that both weights always total 1 and maintain a valid weighted prediction structure.

By systematically evaluating different weight combinations over 200 RMSE-based optimization trials, Table 11 enables the identification of the most effective balance between the two base models, resulting in a more accurate and reliable weighted ensemble prediction.

COLOR-CODED HEATMAP VISUALIZATION

The outbreak for each month is determined using predicted dengue cases, which are compared against a threshold derived from historical dengue data spanning multiple previous years. This approach allows the system to identify potential outbreak periods even when actual case data is unavailable, relying instead on forecasts generated from historical trends and environmental features.

By calculating a monthly threshold from past data, the method accounts for typical seasonal variations in dengue incidence, capturing periods of expected high transmission. The threshold incorporates both the average number of cases and the variability observed in previous years, ensuring that unusually high predictions are flagged as potential outbreaks. The threshold for each month is calculated using the formula:

$$\text{Threshold}_m = \text{Mean}_m + 2 \times \text{StdDev}_m$$

An outbreak is considered to occur when the predicted cases for month mmm exceed the threshold:

$$\text{Predicted Cases}_m \geq \text{Threshold}_m$$

To assess prediction accuracy at the locality–month level, a color-coded heatmap was implemented. The accuracy metric used is the percentage error between actual and predicted dengue cases, calculated as:

$$\text{Percent Error} = \frac{|Actual - Predicted|}{Actual} \times 100$$

When a month is classified as an outbreak, the cell color in the heatmap indicates the prediction accuracy category, as presented in Table 12.

Table 12. Prediction Accuracy Classification for Outbreak Evaluation

Accuracy Classification	Percentage Error Condition	Color
Low error	≤ 10%	Green (#81EE5A)
Moderate error	≤ 30%	Yellow (#FFD700)
High error	> 30%	Red (#E63D34)

Based on the classification by Lewis (1982) and supported by Montaño et al. (2013), forecast accuracy can be categorized using Absolute Percentage Error (APE) as follows: low error (≤10%), moderate error (11–30%), and high error (>30%), with corresponding color codes of green, yellow, and red to facilitate visual interpretation, as shown in Table 12. This adjustment from the original thresholds (<10% highly accurate, 10–20% good, 20–50% reasonable, >50% inaccurate) was made to simplify the classification into three clear categories, which enhances clarity and provides a more practical framework for visualizing and comparing forecasting performance across different localities.

MODEL EVALUATION

To assess the performance of the developed forecasting model, this study employs standard regression evaluation metrics—Mean Absolute Error

(MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). These metrics quantify the accuracy of the model's predictions by measuring the deviation between predicted and actual dengue case values. MAE captures the average magnitude of errors, RMSE emphasizes larger errors by penalizing them more heavily, and R^2 indicates how well the model explains the variability in the dataset. Together, these metrics assess the model's accuracy and generalizability in dengue outbreak forecasting.

Mean Absolute Error (MAE)

Mean Absolute Error (MAE) quantifies the average magnitude of prediction errors in a model. It is calculated by taking the absolute value of the difference between each predicted value and the actual value, then averaging those differences across all data points. Because MAE does not square the errors, it treats all errors equally, whether they are small or large. This makes MAE an intuitive and easy-to-interpret measure of model accuracy, especially when you want a straightforward sense of how far off predictions are from actual outcomes on average. A lower MAE indicates better model performance, as it means the predictions are closer to the actual values on average.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} | y_i - \hat{y}_i |$$

Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) measure the average magnitude of prediction errors, with larger errors receiving more weight due to squaring. It is calculated as the square root of the average of the squared differences between actual and predicted values. RMSE has the same unit as the target variable, making it interpretable in real-world terms. A lower RMSE indicates better

model performance, while a higher RMSE suggests that predictions deviate more from actual values. Because it penalizes larger errors more severely, RMSE is especially useful when large deviations are particularly undesirable.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Coefficient of Determi

Coefficient of Determination (R^2), or R-squared, measures how well the regression predictions approximate the actual data values. It quantifies the proportion of variance in the dependent variable that is explained by the independent variables in the model. R^2 is calculated using two components: the Residual Sum of Squares (RSS) and the Total Sum of Squares (TSS). RSS measures the total squared differences between predicted and actual values, while TSS measures the total squared differences between actual values and their mean. A lower RSS relative to TSS indicates better model performance.

$$R^2 = 1 - \frac{RSS}{TSS}$$

ETHICAL CONSIDERATION

This study followed proper ethical guidelines to make sure the data was used in a responsible way. The data included records of dengue cases from 2016 to 2020, along with related environmental and population information from the Luzon. To get the data, the researchers sent a formal request letter or email to the data sources, which was signed by their thesis adviser. The letter clearly explained the purpose of the request and stated that the data would only be used for academic purposes, not for personal or business use. The data was kept safe and was only accessed by the researchers. The researchers also

made sure that the model was fair and did not show any bias. No experiments were done on people. The goal of this study is to help the community by providing useful information to support public health and help in controlling dengue.

CHAPTER IV

RESULTS AND DISCUSSION

Dengue fever continues to pose a significant public health challenge in the Philippines, largely due to the country's geographic location within a tropical zone. It experiences high temperatures and humidity throughout the year, which provide favorable conditions for mosquito breeding and virus transmission. The seasonal and environmental variability, such as fluctuations in temperature, rainfall, and humidity, further increases the risk of dengue outbreaks.

As such, the development of accurate forecasting methods is critical for anticipating potential surges in dengue cases. Reliable forecasts enable public health authorities to allocate resources efficiently, implement early interventions, and design targeted control strategies to minimize the impact of the disease on affected communities. Predictive modeling using an ensemble of Random Forest and XGBoost through a Weighted Average approach serves not only as a tool for risk assessment but also as a proactive strategy for disease prevention and outbreak mitigation.

This chapter presents the results obtained from the implementation and evaluation of the dengue outbreak forecasting model using the Weighted Average ensemble of Random Forest and XGBoost regression. It provides a detailed discussion of the outcomes of spatial analysis and predictive performance based on Epidemiological, Demographic, and Meteorological Data. The result will be coming from the integration of both spatial analysis outputs and ensemble regression results to deliver a comprehensive

interpretation of outbreak risks, which are visually presented through a color-coded heatmap.

Spatially analyzed dengue outbreak patterns using integrated multi-source datasets and applied Getis-Ord Gi* and DBSCAN to identify statistically significant hotspot areas

This section presents the results of the spatial analysis conducted to identify dengue outbreak patterns. The analysis was performed by integrating multi-source datasets, including epidemiological data (dengue cases), demographic data (population number and population density), and meteorological data (rainfall, relative humidity, max/min temperature, wind speed, and wind direction). This integrated dataset was managed and processed using PostgreSQL with its PostGIS spatial extension. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Getis-Ord Gi*, were applied to identify spatial clusters and statistically significant hotspot areas.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

The application of the DBSCAN algorithm successfully partitioned the localities into 15 distinct clusters. A pronounced concentration of localities is observed in Cluster 3, which emerges as the largest group. This cluster comprises cities within the National Capital Region (NCR) along with provinces from the adjacent CALABARZON and Central Luzon regions, indicating strong spatial similarity among these areas. The second largest grouping is Cluster 6, consisting of the neighboring localities of Benguet, Baguio, and La Union. In contrast, the remaining clusters are predominantly composed of single

localities, suggesting that these areas exhibit distinct spatial characteristics that differentiate them from the larger cluster formations.

Table 13 presents the size of each cluster. Cluster 3 is the most significant, with a size of 7,540. Following this, Cluster 6 is the next largest with a size of 780. The remaining clusters show a distinct pattern, with most having a size of 260. Clusters 5 and 14 are an exception to this, each containing 520. This cluster highlights another significant grouping of dengue cases, though smaller in extent compared to Cluster 3.

Table 13. Cluster ID and Size

Cluster ID	Cluster Size
Cluster 1	260
Cluster 3	7,540
Cluster 4	260
Cluster 5	520
Cluster 6	780
Cluster 7	260
Cluster 8	260
Cluster 9	260
Cluster 10	260
Cluster 11	260
Cluster 12	260
Cluster 13	260
Cluster 14	520
Cluster 15	260

To evaluate the overall clustering of dengue cases across the entire study area, the Global Moran's I statistic was calculated. This statistic measures spatial autocorrelation, the degree to which locations with high (or low) case counts are grouped together in space. A Global Moran's I value near +1

indicates strong clustering, while a value near 0 indicates a random spatial distribution. Statistical significance was assessed using a permutation test with 999 iterations.

Table 14. Global Spatial Autocorrelation

Statistic	Value	Expected	Variance	z-score	p-value
Morans I (cases)	-0.0002	-0.0213	0.0011	0.6135	0.236

Table 14 yield a Moran's I index of -0.0002, which is very close to the expected value of -0.0213 for a random distribution. The calculated z-score of 0.6135 and p-value of 0.236 indicate that the result is not statistically significant at the 0.05 level. This suggests that on a global scale, the distribution of dengue cases does not exhibit a uniform clustering pattern across the entire region. However, this lack of *global* autocorrelation does not preclude the existence of specific *local* hotspots, which are examined in the subsequent Getis-Ord Gi* analysis.

Table 15 summarizes the characteristics of the identified clusters, including the number of points, the percentage of total dataset points they represent, their geometric centroid, and their spatial footprint. Together, these attributes enable systematic comparison of cluster size, density, and spatial distribution. The number of points and corresponding percentage quantify the relative dominance of each cluster within the dataset, while the geometric centroid provides a representative spatial location that facilitates regional interpretation. The spatial footprint reflects the geographic extent and dispersion of each cluster, allowing distinctions between compact, densely

grouped clusters and those that are more spatially spread. Collectively, these measures support a clearer understanding of spatial heterogeneity and the underlying clustering structure across localities.

Table 15. DBSCAN clustering summary

Cluster ID	n_points	% of total points	Centroid (lat,lon)	Spatial extent (km²)	Notes
1	260	2.08	(13.21667, 123.55000)	0.00	Single location
3	7540	60.42	(14.62559, 120.93725)	12668.3	Multi-location cluster
4	520	4.17	(16.16540, 121.62671)	0.00	Single location
5	520	4.17	(18.14333, 120.97000)	0.00	Single location
6	780	6.25	(16.50186, 120.56412)	209.67	Multi-location cluster
8	260	2.08	(17.58000, 120.80000)	0.00	Single location
9	260	2.08	(17.58000, 120.80000)	0.00	Single location
10	260	2.08	(12.16667, 123.58333)	0.00	Single location
11	260	2.08	(13.83333, 124.25000)	0.00	Single location
12	260	2.08	(14.16667, 122.75000)	0.00	Single location
13	260	2.08	(12.83333, 123.91667)	0.00	Single location
14	520	4.17	(15.97992, 120.33376)	0.00	Single location
15	260	2.08	(15.58333, 121.00000)	0.00	Single location

The analysis revealed 15 distinct clusters across the study region. Cluster 3 is the most significant, containing 60.42% of all data points and

covering a vast spatial extent of approximately 12,668 km², suggesting a massive, contiguous region of high dengue activity likely spanning multiple provinces. In contrast, most other clusters like Clusters 1, 2, and 4 have a spatial extent of 0 km², indicating that they represent isolated municipalities where cases are geocoded to a single centroid point rather than spread across a wider area.

Getis Ord Gi*

The Getis-Ord Gi* revealed spatial clustering patterns of dengue incidence. From the dataset of 12,480, Figure 3 illustrates the results identified 146 hotspots (areas of significantly elevated incidence) and 373 coldspots (areas with significantly fewer cases than expected). The vast majority were classified as not significant, exhibiting no statistically significant clustering tendency and suggesting a random spatial distribution.

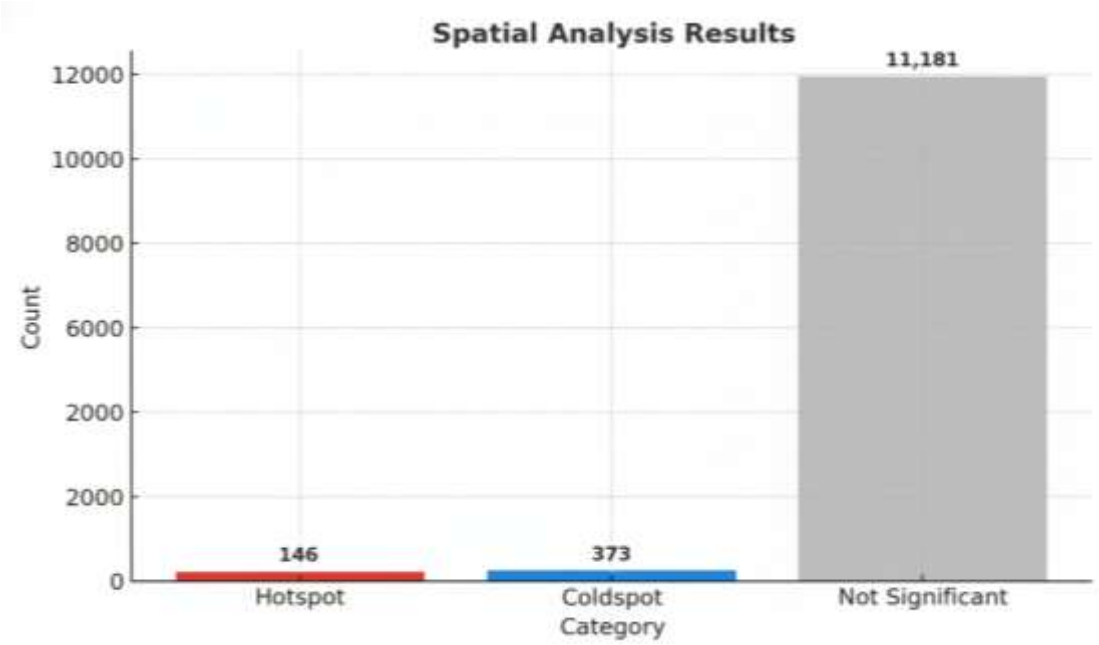


Figure 3. Results of Getis Ord Gi*

These findings demonstrate that dengue outbreaks follow distinct spatial patterns when multi-source data are analyzed together. The integration of demographic and meteorological variables strengthened the detection of meaningful hotspots and reduced noise, ensuring that the identified high-risk areas are reliable. The Getis-Ord Gi* statistic was employed to identify statistically significant spatial clusters of dengue cases. This analysis distinguishes between random spatial distributions and significant 'hotspots' areas where high numbers of dengue cases

Table 16. Getis-Ord Gi Top Hotspots

Admin unit ID	Centroid (lat,lon)	Gi* z-score	p-value	Adjusted p-value (FDR)	Hotspot? (yes/no)	Date
Batangas	(13.91468, 121.08676)	6.8079	0.001	0.035404	hotspot	2016-08-07
Batangas	(13.91468, 121.08676)	6.7532	0.001	0.035404	hotspot	2016-07-31
Batangas	(13.91468, 121.08676)	6.7489	0.001	0.035404	hotspot	2018-09-09
Batangas	(13.91468, 121.08676)	6.7065	0.002	0.035404	hotspot	2016-07-24
Batangas	(13.91468, 121.08676)	6.677	0.002	0.035404	hotspot	2017-09-10
Batangas	(13.91468, 121.08676)	6.677	0.002	0.035404	hotspot	2019-09-15
Batangas	(13.91468, 121.08676)	6.6769	0.002	0.035404	hotspot	2020-09-20
Batangas	(13.91468, 121.08676)	6.6769	0.002	0.035404	hotspot	2020-09-06
Batangas	(13.91468, 121.08676)	6.6769	0.002	0.035404	hotspot	2020-08-23
Batangas	(13.91468, 121.08676)	6.6769	0.002	0.035404	hotspot	2020-08-16

Table 16 highlights specific localities, such as Batangas, that exhibited the most intense clustering of dengue cases during the study period. The Gi z-score* measures the intensity of the clustering; a higher positive z-score indicates a more intense hotspot. The Adjusted p-value (FDR) confirms that these patterns are statistically significant ($p < 0.05$) even after correcting for the multiple spatial tests performed. These results pinpoint critical areas where vector control interventions should be prioritized due to the non-random, high concentration of cases.

Developed Weighted Average Ensemble Model Using Random Forest and XGBoost Based on Combined Epidemiological, Meteorological, Demographic, and Spatial Analysis Inputs

The developed weighted averaging ensemble model combines Random Forest and XGBoost to enhance dengue outbreak prediction accuracy. Both models were individually optimized using Bayesian Optimization to obtain their best hyperparameter configurations. After optimization, their predictions were integrated through a weighted averaging approach, where Optuna determined the optimal weight distribution to minimize prediction error. This ensemble leverages the stability of Random Forest and the precision of XGBoost, producing a balanced and reliable predictive output for dengue outbreak forecasting using epidemiological, meteorological, demographic, and spatial analysis data.

Random Forest (Bayesian Optimization)

The Random Forest model was employed in this study to predict dengue cases using a comprehensive set of integrated inputs encompassing epidemiological, meteorological, demographic, and spatial variables. As an

ensemble learning method based on multiple decision trees, Random Forest was designed to improve predictive accuracy and control overfitting by averaging the outputs of numerous trees trained on different subsets of the data. Bayesian Optimization was applied to fine-tune the model's hyperparameters, ensuring optimal depth, number of estimators, etc. to maximize generalization performance. This approach enabled the Random Forest to effectively capture complex interactions among variables influencing dengue transmission, providing a stable and reliable baseline model for outbreak prediction.

Table 17. Hyperparameter of Random Forest

Hyperparameter	**Value**
n_estimators	562
max_depth	14
min_samples_split	7
min_samples_leaf	4
max_features	None
bootstrap	True

Table 17 shows the best-performing hyperparameter configuration obtained from the Random Forest optimization using Bayesian search in Optuna. The model achieved its strongest performance using 562 estimators, providing a sufficient number of trees to enhance predictive accuracy without excessive computation. The maximum depth was optimized to 14, allowing the trees to capture complex patterns while still managing overfitting. The min_samples_split value of 7 and min_samples_leaf value of 4 introduce

necessary constraints on node splitting, supporting better generalization on unseen data.

Also, the max_features parameter was set to None, meaning all features were considered when building each split, maximizing learning capacity. The inclusion of bootstrap=True preserves sampling with replacement, which increases tree diversity and strengthens ensemble robustness. The optimized hyperparameters in Table 17 ensure a well-tuned Random Forest model that performs reliably as a base learner in the weighted average ensemble.

XGBoost (Bayesian Optimization)

The XGBoost model was developed to capture the intricate nonlinear relationships between dengue cases and a diverse range of predictor variables, including climatic, demographic, and spatial factors. Known for its gradient boosting framework, XGBoost builds decision trees sequentially, where each tree corrects the errors of its predecessors, resulting in a strong predictive model. This sequential learning process enables the model to reduce bias and variance simultaneously, improving overall generalization performance. As a result, XGBoost is well suited for complex epidemiological data where relationships among variables are rarely linear or independent.

Bayesian Optimization was utilized to determine the most effective combination of hyperparameters, such as learning rate, tree depth, and regularization terms, to balance model complexity and accuracy. This process enhanced the model's ability to identify subtle, high-order dependencies among predictors, making XGBoost particularly effective in handling the heterogeneity and variability present in dengue case data. By systematically exploring the hyperparameter space, Bayesian Optimization reduced the risk of overfitting

while improving predictive stability. This approach also ensured efficient model tuning compared to exhaustive search methods, which can be computationally expensive.

Table 18. Hyperparameter of XGBoost

Hyperparameter	Value
n_estimators	105
max_depth	4
learning_rate	0.07558836682599704
subsample	0.9792114669799131
colsample_bytree	0.5647576463467691
gamma	9.709730099421618
reg_alpha	2.9684473356688663
reg_lambda	4.426489107764399

Table 18 presents the optimal hyperparameter configuration for the XGBoost model obtained through Bayesian optimization using Optuna. With 105 boosting rounds, the model avoids excessive complexity while retaining strong predictive performance. A maximum depth of 4 limits tree growth, reducing the likelihood of overfitting. The learning rate of approximately 0.0756 ensures gradual but stable learning from each boosting iteration.

The subsample value of about 0.9792 means most of the data is used each round, balancing performance and variability during training.The colsample_bytree value of around 0.5648 restricts the number of features used per tree, helping introduce diversity between trees and prevent reliance on specific features. A gamma value of approximately 9.71 imposes a strong

constraint on further splitting, encouraging only meaningful improvements in model structure.

Meanwhile, L1 (reg_alpha ≈ 2.97) and L2 (reg_lambda ≈ 4.43) regularization penalties control weight complexity, further improving generalization. With this optimized parameter set in Table 18, the XGBoost model becomes more robust and accurate, making it a strong base model for the weighted average ensemble.

Weighted Average Ensemble

Table 19. Weights of Base Model

Base Model	Weight
Random Forest (w_rf)	0.1121
XGBoost (w_xgb)	0.8879

Table 19 displays the optimal weight distribution obtained from the Optuna-based weight tuning for the weighted average ensemble model. The Random Forest model received a weight of 0.1121, indicating that it contributes a smaller portion to the final ensemble prediction. In contrast, the XGBoost model was assigned a much higher weight of 0.8879, showing that it provides the dominant influence on the ensemble's output.

This weight allocation reflects the stronger predictive performance of the XGBoost model during optimization, where Root Mean Squared Error (RMSE) was used as the basis for selecting the best-performing weight combination. As shown in Table 19, these optimized weights ensure that the ensemble takes

advantage of both models while prioritizing the one that delivers higher accuracy, resulting in an improved and more reliable prediction performance.

Evaluation of the Model's Predictive Accuracy Using Regression Metrics (MAE, RMSE, R^2) and Visualization of Forecasted Dengue Outbreak Risks through a Color-Coded Heatmap

This section evaluates the model's predictive accuracy using three standard regression metrics—Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2)—to assess both the precision of predictions and the model's ability to explain variations in dengue case counts. Lower MAE and RMSE values indicate that the forecasted dengue cases are close to the actual records, while a higher R^2 value reflects strong model reliability in capturing outbreak patterns. In addition to numerical evaluation, a color-coded heatmap is used to visualize forecasted dengue outbreak risks across months and localities, providing clear insights for decision-making.

Table 20. Model Predictive Accuracy Result

Metric	Random Forest	XGBoost	Weighted Ensemble
R-Squared	0.921752	0.926351	0.926423
MAE	4.648624	4.628130	4.609461
RMSE	8.042851	7.802892	7.799064

Table 20 compares the predictive performance of the Random Forest, XGBoost, and Weighted Ensemble models using R^2, MAE, and RMSE as evaluation metrics. These values measure how closely the forecasted dengue cases match the actual recorded cases. Higher R^2 values indicate that a model

explains more variation in dengue cases, while lower MAE and RMSE values indicate fewer prediction errors. Both base models demonstrate strong forecasting performance, but each contributes differently to the overall predictive ability.

Between the two base models, XGBoost performs better than Random Forest across all metrics. However, the Weighted Ensemble—by combining both models—provides the strongest results. For R^2, the Weighted Average Ensemble improves performance by 0.004671 over Random Forest (0.926423 vs. 0.921752) and by 0.000072 over XGBoost (0.926423 vs. 0.926351). In terms of MAE, the Weighted Ensemble reduces error by 0.039163 compared to Random Forest and by 0.018669 compared to XGBoost. Meanwhile, RMSE is lowered by 0.243787 relative to Random Forest and by 0.003828 compared to XGBoost. These differences, while small in numeric scale, reflect a measurable improvement in predictive precision.

The developed Weighted Ensemble model, integrating Random Forest and XGBoost, achieved an R^2 of 0.926, MAE of 4.61, and RMSE of 7.80, outperforming several previous dengue forecasting studies. In comparison, Olmoguez et al. (2019) attained only 73% accuracy ($R^2 = 0.73$) using Random Forest in Iligan City, indicating limited explanatory power for dengue variability based solely on climatic factors. Similarly, Patil and Pandya (2021) reported a maximum R^2 below 0.90 in forecasting dengue hotspots across Maharashtra using regression and time series models.

More recently, Mehta and Patel (2024) employed an LSTM-based approach for Gujarat, producing an R^2 of 0.84. Compared to these findings, the ensemble model presented in this study demonstrates superior predictive

accuracy and lower error rates, suggesting that combining Random Forest and XGBoost effectively captures complex nonlinear relationships between demographic and environmental predictors. This enhanced performance validates the model's robustness and reliability for regional dengue forecasting applications.

Overall, the Weighted Average Ensemble outperforms both base models by leveraging their individual strengths, XGBoost's superior learning of complex nonlinear patterns and Random Forest's robust tree-based decision structure. As shown in Table 20, this combination reduces bias, lowers variance, and achieves the most accurate dengue case forecasts among the tested models. It is normal for the improvement to appear small because the base models are already highly optimized and performing well individually, leaving less space for a large numerical jump in accuracy.

Table 21. Test Set Performance (Aggregated)

Metric	**Value**	**95% CI (bootstrap)**	**Notes**
MAE	4.748553	[4.6306, 4.8788]	Lower is better
RMSE	7.869673	[7.5810, 8.2070]	Lower is better
R^2	0.943952	[0.9390, 0.9484]	Closer to 1 indicates better fit
Poisson deviance	2.695962	[2.5554, 2.8534]	Lower indicates better count prediction

Table 21 summarizes the predictive performance of the final model, demonstrating strong accuracy and stability across evaluation metrics. The low MAE (4.75) and RMSE (7.87) indicate small prediction errors, while the high R^2

value (0.94) shows that most variability in dengue cases is well explained. The low Poisson deviance (2.70) confirms suitability for count-based data. Narrow 95% bootstrap confidence intervals further indicate robust and consistent performance, supporting the model's reliability for short-term dengue forecasting and outbreak risk assessment.

Table 22. Feature Importance (Top 15)

Rank	Feature	RF importance (mean)	XGB importance (gain)	Permutation importance (RF)	SHAP mean abs
1	cases_lag_1	0.4022	0.2625	9.5321	8.9634
2	cases_ema_alpha_0.5	0.4660	0.2306	19.8499	8.5514
3	cases_roll_max_3w	0.0426	0.3230	0.3762	3.3406
4	cases_lag_2	0.0084	0.0366	4.0427	2.8095
5	week_cos	0.0119	0.0064	2.1597	1.1821
6	week_sin	0.0045	0.0047	0.7448	1.1355
7	population	0.0025	0.0041	0.8523	1.1103
8	cases_roll_mean_3w	0.0043	0.0334	0.3534	0.9390
9	cases_lag_4	0.0032	0.0047	0.4963	0.8993
10	day	0.0027	0.0042	0.2645	0.3496
11	month	0.0018	0.0030	0.2149	0.3292
12	month_sin	0.0008	0.0043	0.0940	0.2702
13	cases_roll_mean_14w	0.0018	0.0024	0.1889	0.2580
14	cases_roll_std_7w	0.0039	0.0019	0.2839	0.2545
15	year	0.0016	0.0017	0.1851	0.2479

Table 22 summarizes feature importance across Random Forest, XGBoost, permutation importance, and SHAP analyses, showing strong agreement that short-term historical dengue cases are the primary drivers of model predictions. Lagged case variables—particularly cases_lag_1, cases_ema_alpha_0.5, and cases_roll_max_3w—consistently rank highest, indicating that recent case trends and short-window extremes dominate predictive performance.

Seasonal components (week_sin, week_cos) contribute modestly, capturing periodic patterns, while longer lags and rolling statistics add incremental value. Demographic context (population) has a smaller but non-negligible effect, suggesting it refines risk levels rather than drives them. Calendar variables (day, month, year) show limited influence, confirming that temporal structure is better captured through lagged and smoothed case features. Overall, the table indicates a prediction framework driven primarily by recent incidence dynamics, with seasonality and population acting as secondary modifiers.

Table 23. Performance by Area

Region	Actual Cases 2020	Predicted Cases 2020	Threshold	MAE	RMSE	R2
CAR	875	990	49.9558	7.59	10.60	0.58
NCR	6831	7255	509.2275	30.50	44.82	0.94
Region III	12827	12731	538.5709	19.11	26.29	0.96
Region IV-A	6578	6668	384.9046	14.38	20.85	0.97
Region V	924	1012	48.2571	6.76	8.60	0.67
Region I	4245	4270	168.1543	9.48	14.15	0.89

Table 23 presents the model's performance by area for the year 2020, comparing total actual and predicted dengue cases, risk band classification, and error metrics across regions. The model correctly classified all regions as Outbreak, showing full agreement between actual and predicted risk bands relative to the defined thresholds.

Prediction accuracy varies by region, with higher R^2 values observed in densely populated and high-incidence areas such as NCR ($R^2 = 0.94$), Region III ($R^2 = 0.97$), and Region IV-A ($R^2 = 0.97$), indicating strong explanatory power. Regions with lower total case counts, such as CAR and Region V, exhibit higher relative error and lower R^2 values, reflecting greater variability and sensitivity to fluctuations in smaller case volumes. Overall, the results indicate that the model performs consistently well across regions, particularly in high-burden areas, while maintaining correct outbreak risk identification nationwide.

Color-coded Heatmap Visualization

The color-coded heatmap visualization is designed to highlight areas with potential dengue outbreaks. In this heatmap, each color represents both the predicted intensity of cases and the confidence in those predictions. By combining magnitude and reliability in a single view, the heatmap simplifies complex prediction outputs into an interpretable format. This approach supports rapid spatial comparison across regions and time periods. Green indicates regions where the predicted number of cases is highly accurate, suggesting reliable data and low risk of error.

Yellow represents moderate accuracy, signaling that while there may be a potential outbreak, the prediction carries some uncertainty. Red highlights areas with low prediction accuracy, implying that the risk of an outbreak is less

certain or the model's confidence in these predictions is limited. This visual representation allows public health officials and stakeholders to quickly identify regions that require closer monitoring and targeted intervention.



Figure 4. Color-Coded Heatmap Visualization (NCR)

Figure 4 presents the color-coded heatmap of dengue outbreak detection across the National Capital Region (NCR). Each cell represents the predicted number of dengue cases per city and month, with colors indicating prediction confidence levels. Green cells denote highly accurate predictions that align closely with observed outbreaks.

Yellow cells capture general trends with minor deviations, while red cells indicate low accuracy or potential false detections. Quezon City, Pasig City, and Taguig City show strong prediction performance, while Mandaluyong City, Navotas City, and Pasay City display less consistent results, reflecting variations in model reliability across NCR.

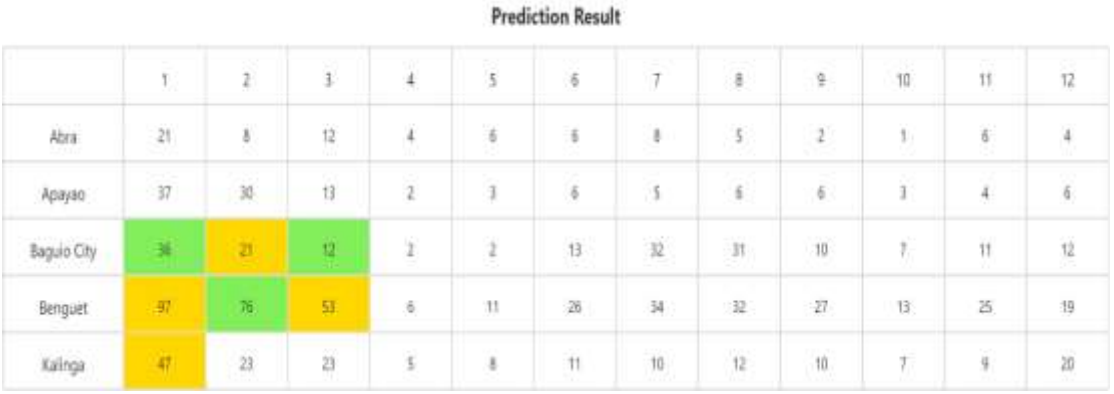


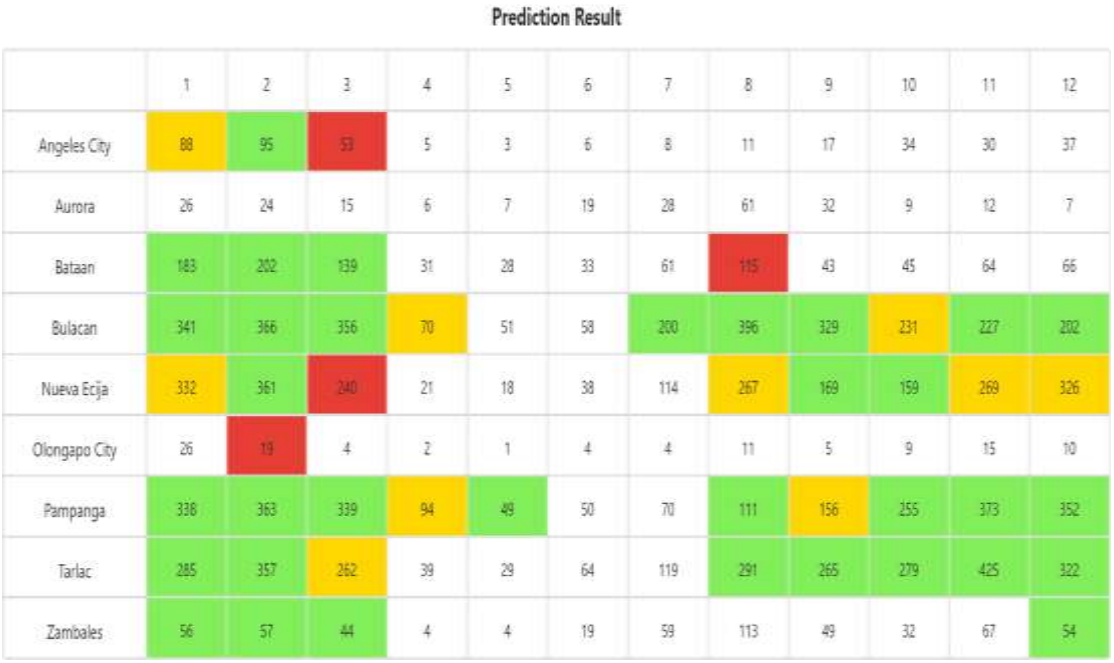
Figure 5. Color-Coded Heatmap Visualization (CAR)

Figure 5 displays the color-coded heatmap for the Cordillera Administrative Region (CAR), showing monthly predicted dengue cases per province. Green cells indicate strong alignment between predicted and observed patterns, while yellow cells show moderate accuracy with slight deviations. No red cells are observed, indicating stable model performance. Baguio City achieved the highest accuracy, followed by Benguet and Kalinga, which maintained consistent moderate results. The findings suggest effective outbreak detection in areas with well-documented data, particularly in urban centers like Baguio City.



Figure 6. Color-Coded Heatmap Visualization (Region I)

Figure 6 presents the heatmap for Region I (Ilocos Region), where each cell corresponds to predicted dengue cases per province and month. Green indicates high accuracy, yellow moderate, and red low accuracy. Pangasinan recorded consistently accurate predictions, while Dagupan City, Ilocos Norte, and La Union showed mixed results with several yellow and red cells. These discrepancies may be due to irregular case trends, indicating that data-rich provinces like Pangasinan support better model performance.



	1	2	3	4	5	6	7	8	9	10	11	12
Angeles City	88	95	53	5	3	6	8	11	17	34	30	37
Aurora	26	24	15	6	7	19	28	61	32	9	12	7
Bataan	183	202	139	31	28	33	61	115	43	45	64	66
Bulacan	341	366	356	70	51	58	200	396	329	231	227	202
Nueva Ecija	332	361	240	21	18	38	114	267	169	159	269	326
Olongapo City	26	19	4	2	1	4	4	11	5	9	15	10
Pampanga	338	363	339	94	49	50	70	111	156	255	373	352
Tarlac	285	357	262	39	29	64	119	291	265	279	425	322
Zambales	56	57	44	4	4	19	59	113	49	32	67	54

Figure 7. Color-Coded Heatmap Visualization (Region III)

Figure 7 shows the heatmap for Region III (Central Luzon), where green marks high accuracy, yellow moderate, and red low accuracy. Bulacan, Pampanga, and Tarlac maintained strong prediction accuracy, while Nueva Ecija and Bataan displayed moderate fluctuations. Angeles City and Olongapo City recorded red cells, suggesting weaker alignment in areas with fewer cases. The results indicate that reliable predictions are achieved in provinces with comprehensive case data.



Figure 8. Color-Coded Heatmap Visualization (Region IV-A)

Figure 8 presents the heatmap for Region IV-A (CALABARZON), where green indicates high accuracy, yellow moderate, and red low accuracy. Batangas, Cavite, Laguna, and Rizal consistently achieved accurate results, reflecting stable reporting and strong data availability. Lucena City displayed moderate accuracy in earlier months, while Cavite showed a few low-accuracy predictions during transitional outbreak periods. The model demonstrates strong reliability in densely populated provinces with robust reporting systems.



Figure 9. Color-Coded Heatmap Visualization (Region V)

Figure 9 shows the heatmap for Region V (Bicol Region), illustrating monthly dengue case predictions per province. Green represents high accuracy, yellow moderate, and red low accuracy. Albay and Sorsogon recorded strong prediction performance, while Camarines Norte and Catanduanes showed weaker alignment. Masbate exhibited variable accuracy across months, likely due to changing dengue case patterns in the area.

Line Graph

The line graph illustrates the trend of cases over the months, comparing actual and predicted values. The horizontal axis represents the months, while the vertical axis shows the number of cases. The red line indicates the actual reported cases, providing a reference for real-world observations, and the blue line represents the predicted cases generated by the model. By visualizing both lines together, the graph allows for an easy comparison between observed and forecasted data, highlighting periods where predictions closely match the actual cases and areas where discrepancies occur. This helps evaluate the accuracy and reliability of the predictive model over time.

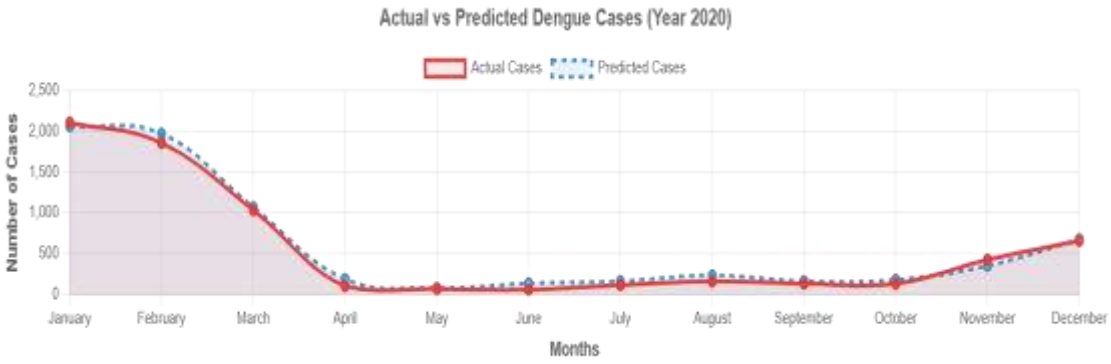


Figure 10. Comparative Line Graph (NCR)

Figure 10 presents a comparative line graph of actual and predicted dengue cases in the National Capital Region (NCR) for the year 2020. The data show a steep decline in dengue incidence from January to April, coinciding with the onset of the dry season and the effects of pandemic-related restrictions that may have reduced mosquito exposure and breeding activity.

The predicted values closely follow the actual cases, particularly during the midyear period when infection levels remained low, indicating strong model reliability during stable transmission phases. Minor discrepancies appear at the

beginning and end of the year, as actual cases were slightly higher in the early months, while the model modestly underestimated the upward trend in November and December.

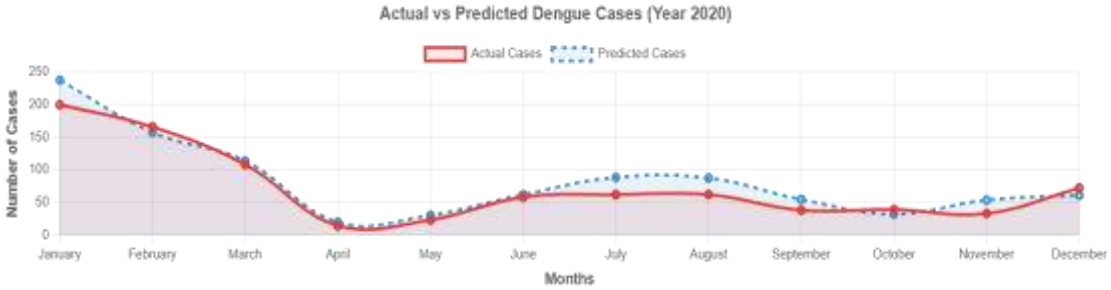


Figure 11. Comparative Line Graph (CAR)

Figure 11 shows the comparison between actual and predicted dengue cases in the Cordillera Administrative Region (CAR) for the year 2020. The trend demonstrates a rapid decrease in dengue cases from January to April, aligning with seasonal patterns were cooler temperatures and reduced rainfall limit mosquito activity. From May onward, both actual and predicted values exhibit a gradual rise, with cases peaking between July and August before declining again toward October. The predicted line follows the actual data closely, reflecting the model's capability to track dengue dynamics within the region. Slight underestimations are visible in the early and later months, indicating that while the model effectively captures the general trend and seasonal variations, it shows less precision during transition periods.

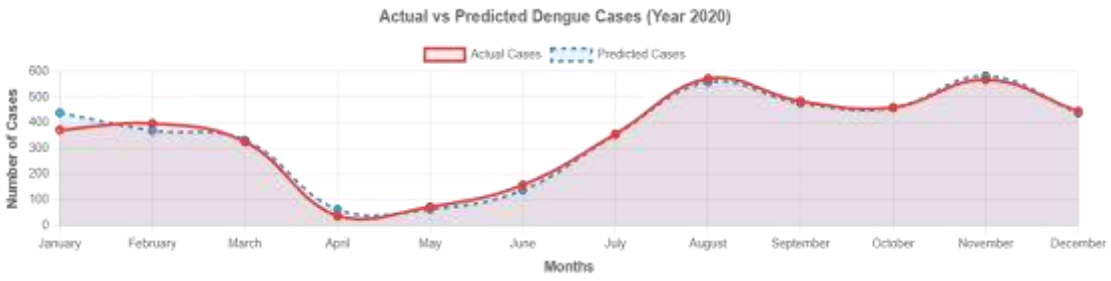


Figure 12. Comparative Line Graph (Region I)

Figure 12 illustrates the comparison between actual and predicted dengue cases in Region I for the year 2020. The data reveal a gradual decline in dengue incidence from January to April, coinciding with the transition into the dry season. Beginning in May, both actual and predicted cases start to increase, peaking around August, which aligns with the wet season when mosquito breeding conditions are more favorable. The pattern then shows a slight decline toward October before another rise in November, suggesting secondary transmission activity later in the year.

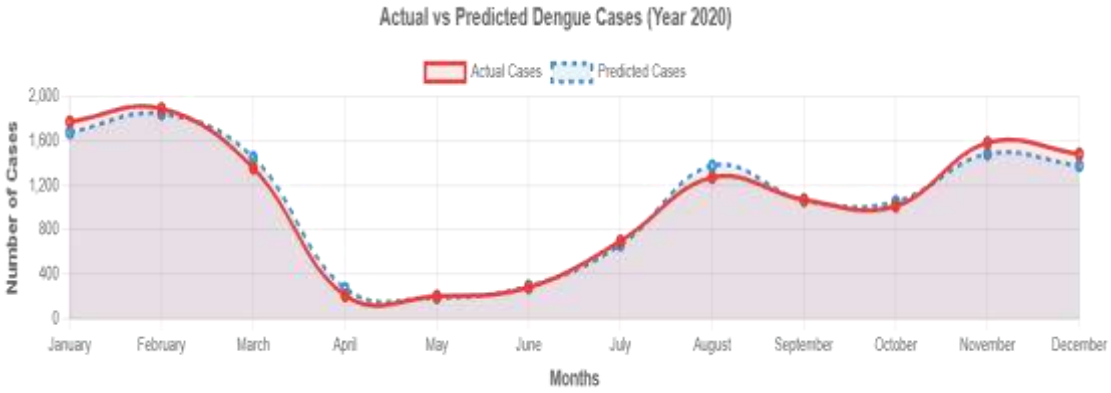


Figure 13. Comparative Line Graph (Region III)

Figure 13 presents the comparison between actual and predicted dengue cases in Region III for the year 2020. The trend shows a high number of dengue cases in the early months, followed by a gradual decrease from January to April, reflecting the seasonal decline as rainfall diminishes.

From May to August, both actual and predicted cases increase sharply, peaking in August when weather conditions are most favorable for mosquito breeding. A temporary decrease follows in September, succeeded by another rise toward November, indicating a secondary transmission period. The predicted values align strongly with the actual data.

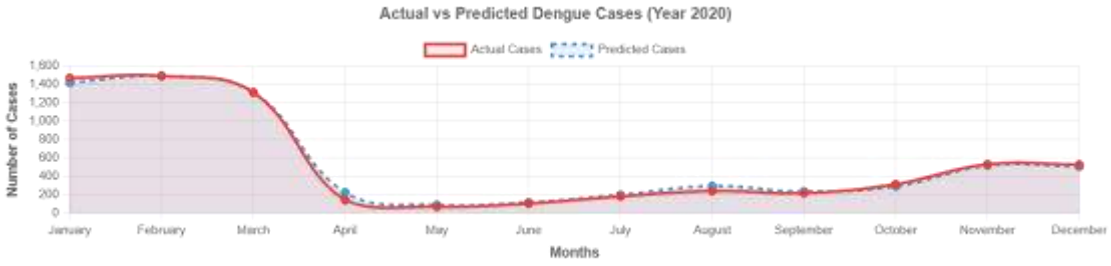


Figure 14. Comparative Line Graph (Region IV-A)

Figure 14 depicts the comparison between actual and predicted dengue cases in Region IV-A for the year 2020. The data show consistently high dengue incidence from January to March, followed by a steep decline beginning in April, coinciding with the transition into the dry season and the implementation of early-year public health interventions.

Between May and September, dengue activity remains relatively low, with both actual and predicted cases maintaining a similar trend, reflecting the model's accuracy during this stable phase. Starting from October, the number of cases begins to rise again, reaching a secondary peak in November before slightly tapering in December. The predicted line follows the actual pattern closely across all months, demonstrating the model's ability to capture the region's seasonal dengue fluctuations with minimal deviation.

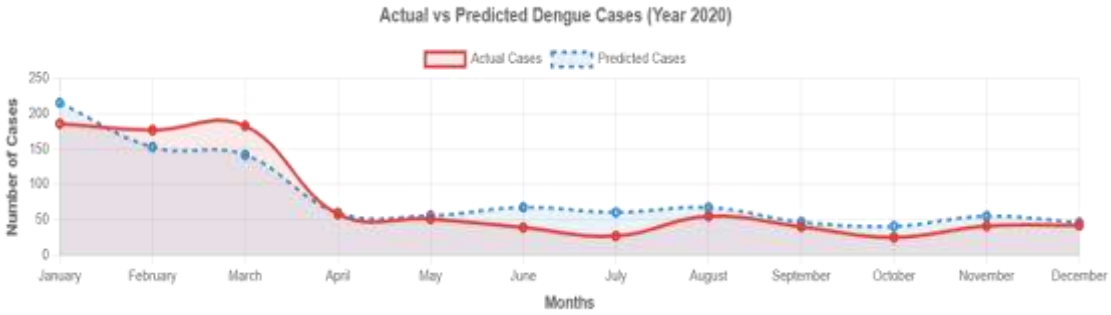


Figure 15. Comparative Line Graph (Region V)

Figure 15 illustrates the comparison between actual and predicted dengue cases in Region V for the year 2020. The trend reveals a moderate level of dengue incidence during the first quarter, followed by a sharp decrease beginning in April, consistent with the onset of the dry season and reduced rainfall. From May to September, dengue cases remain relatively low and stable, with both actual and predicted values showing close alignment, indicating accurate model performance during low transmission periods.

A slight increase is observed from June to August, coinciding with the rainy months that favor mosquito breeding, after which the numbers stabilize through the year's end. The predicted line effectively mirrors the overall movement of the actual data, demonstrating reliable forecasting of seasonal fluctuations.

Statistical Test

Table 24. Summary of Statistical Comparison between Actual and Predicted Dengue Cases

Data Type	N	Mean	Median	SD	SE
Actual Cases	500	12.124	3	22.67706159	1.014149025
Predicted	500	12.354	4	21.39130337	0.956648169

Table 24 presents a detailed comparison of descriptive statistics between actual reported dengue cases and predicted cases generated by the forecasting model. The dataset includes 500 inputs for both actual and predicted values. The actual cases have a mean of 12.12 and a median of 3, indicating that most periods have relatively low case counts, while certain periods show higher counts, reflected in the large standard deviation (SD) of

22.68. The standard error (SE) of 1.01 provides an indication of the precision of the mean estimate.

Predicted cases have a slightly higher mean of 12.35 and a median of 4, with a lower SD of 21.39 and SE of 0.96. These statistics suggest that the model closely follows the trends observed in actual cases while exhibiting slightly less variability. The close alignment between the means and medians of actual and predicted data shows that the model captures the central tendency of dengue cases effectively.

The high SD values in both actual and predicted cases reflect the fluctuations common in dengue outbreaks, demonstrating that while the model smooths some of the extreme peaks, it maintains an accurate representation of trends. These descriptive statistics provide a basis for further inferential analysis, such as paired t-tests, to determine whether differences between actual and predicted cases are statistically significant. The results indicate that the predictive model can be considered reliable for estimating dengue case patterns and supporting public health decision-making and outbreak preparedness.

Table 25. Paired T-Test Results Comparing Actual and Predicted Dengue Cases

Paired T-Test	Statistic	DF	p	Mean Difference	Effect Size (Cohen's d)
Actual and Predicted Dengue Cases	-0.90060697	499	0.36823 1765	-0.23	-0.040276368

Table 25 shows the results of a paired t-test conducted to evaluate the difference between actual and predicted dengue cases. The paired t-test is

appropriate because it compares two related samples: the actual cases and the model-generated predictions for the same periods. The test produced a t-statistic of -0.901 with 499 degrees of freedom (DF) and a p-value of 0.368. The mean difference between actual and predicted cases is -0.23, indicating that, on average, predicted values are slightly higher than the actual cases, but the difference is minimal.

The effect size, measured using Cohen's d, is -0.04, suggesting a negligible practical difference between actual and predicted cases. Since the p-value is greater than the significance level of 0.05, the null hypothesis that there is no significant difference between the two sets of values cannot be rejected. This result indicates that the predictive model closely approximates the actual dengue cases, producing estimates that are statistically similar to observed data. These findings provide strong evidence that the forecasting model is reliable and effective for estimating dengue case patterns, supporting its use for public health monitoring and planning.

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMENDATIONS

SUMMARY OF THE STUDY

This study focused on developing a model to forecast dengue outbreaks, which continue to be a serious public health issue. The research used data from 2016 to 2020, combining three different types of data, dengue case records from the Department of Health (DOH), population data from the Philippine Statistics Authority (PSA), and weather information from PAG-ASA. The study was using spatial analysis to find patterns. Using DBSCAN, the research identified 15 distinct spatial clusters of high-risk localities. The largest of these was Cluster 3, a major high-risk zone that includes the National Capital Region (NCR) and many surrounding provinces in CALABARZON and Central Luzon. The Getis-Ord Gi* analysis also successfully pinpointed 146 specific hotspots and 375 coldspots.

To make the forecast, the study used an ensemble method called a Weighted Average Ensemble. This method combined two machine learning models, Random Forest and XGBoost. The final ensemble model gave more importance to the XGBoost model (0.8879 weight) and less to the Random Forest model (0.1121 weight) to get the best results. This combined model showed very high accuracy. It performed better than either of the individual models. The final ensemble achieved an R-Squared value of 0.926423 and had the lowest error scores (MAE: 4.609461 and RMSE: 7.799064). The results were then used to create color-coded heatmaps, providing a simple visual tool to show which areas and months are at the highest risk for a dengue outbreak.

CONCLUSIONS

1. **To spatially analyze dengue outbreak patterns by integrating multi-source data comprising Epidemiological, Demographic, and Meteorological datasets using PostgreSQL/PostGIS, and by applying spatial clustering algorithms such as Getis-Ord Gi* and DBSCAN to identify statistically significant hotspot areas.**

The study successfully met its first objective by spatially analyzing dengue outbreak patterns. Through the integration of epidemiological, demographic, and meteorological data, and the application of DBSCAN and Getis-Ord Gi* analysis, it was concluded that dengue patterns are not random but are highly concentrated in specific areas. The analysis successfully identified 15 distinct spatial clusters and 146 statistically significant hotspots, with "Cluster 3" (covering NCR and parts of CALABARZON and Central Luzon) being confirmed as the largest and most critical high-risk zone.

2. **To develop weighted average ensemble that integrates Random Forest and XGBoost using the combined epidemiological, meteorological, and demographic datasets, along with spatial analysis results.**

The second objective, to develop a superior forecasting model, was successfully achieved. The study concludes that the Weighted Average Ensemble model, which integrates Random Forest and XGBoost, is a highly effective and robust method for this problem. By combining the strengths of both models (with an 0.8879 weight for XGBoost and 0.1121 for Random Forest), the ensemble was proven to be more accurate and reliable for forecasting dengue cases than using either of the individual models on their own.

3. **To evaluate the model's predictive accuracy using regression metrics (MAE, RMSE, R2) and to visualize the forecasted dengue outbreak risks through a color-coded heatmap, to support geographically targeted public health interventions.**

The third objective was met by evaluating the model and creating practical visual tools. The ensemble model demonstrated very high predictive accuracy, achieving an R-Squared value of 0.926423 and the lowest error scores (MAE: 4.609461, RMSE: 7.799064). The study therefore concludes that the resulting color-coded heatmaps and cluster maps are practical and effective tools for visualizing risk. They successfully translate complex model data into a simple format that can be directly used by public health officials and LGUs to support timely and evidence-based decisions.

This study presents a forecasting approach by fusing multi-source data which many studies lack. Its contribution is it integrates the results of spatial analysis (DBSCAN and Getis-Ord Gi*) as new model features. It developed a Weighted Average Ensemble of Random Forest and XGBoost, which was proven to be better than the base models alone. This integrated model provides a more reliable tool for public health by capturing outbreak and visualizing risk in a color-coded heatmap.

RECOMMENDATIONS

The recommendations below aim to inform both future research and the application of dengue outbreak prediction in practice:

1. Public health offices and local government units can utilize predictive models to identify high-risk areas more effectively. This allows for targeted interventions, including mosquito control operations, focused

public awareness campaigns, and the strategic allocation of medical resources, which can significantly reduce the impact of dengue outbreaks before peak seasons.

2. Future studies should aim to incorporate additional data types that were not available in this research, such as mosquito population counts, patterns of human mobility, and local sanitation or water storage practices. Including these factors can improve model accuracy and provide a more detailed understanding of how dengue spreads across different regions.
3. Developing a real-time forecasting system that connects to live data from agencies like PAG-ASA, PSA, and DOH would allow continuous monitoring of dengue risk. This system could provide daily or weekly updates, enabling public health officials to take timely and proactive measures to prevent outbreaks.
4. Exploring alternative modeling approaches, including deep learning techniques, hybrid models, or other advanced machine learning algorithms, may further enhance prediction accuracy. Comparing multiple models can help determine which methods perform best for long-term forecasting and for different regional conditions.
5. Engaging communities through education programs on mosquito control, personal protection measures, and proper hygiene practices is essential for effective dengue prevention. Active participation from residents ensures that interventions are applied consistently and reinforces the overall effectiveness of public health strategies.

6. Future work could directly extend this study's novelty using spatial analysis results as new predictive features for the machine learning model. This methodology could be applied to other public health issues that have strong geographic patterns, such as tracking malnutrition, forecasting malaria outbreaks, or modeling the spread of other infectious diseases. This same technique could also be tested on more detailed data, like at the barangay level, to create even more precise hotspot features. Furthermore, this data fusion method could be used to integrate other novel datasets, such as human mobility patterns or local water sanitation reports, to further improve the model's accuracy.

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APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the thesis entitled:

"SPATIAL ANALYSIS AND MULTI-SOURCE DATA FUSION FOR DENGUE OUTBREAK FORECASTING USING AN ENSEMBLE OF RANDOM FOREST AND XGBOOST WITH HEATMAP VISUALIZATION "

WINCHESTER B. NUÑEZ and NEAL LOUIS V. VILLAMOR have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

CRISANTO F. GULAY
Thesis Adviser
October 30, 2025

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

APPROVAL SHEET

The Thesis entitled **"SPATIAL ANALYSIS AND MULTI-SOURCE DATA FUSION FOR DENGUE OUTBREAK FORECASTING USING AN ENSEMBLE OF RANDOM FOREST AND XGBOOST WITH HEATMAP VISUALIZATION"** prepared and submitted by Winchester B. Nunez and Neal Louis V. Villamor in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

CRISANTO F. GULAY
Thesis Project Adviser

APPENDIX B

Data Acquisition Request Letter

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

July 01, 2025

Dr. Nathaniel T. Servando
PAGASA Administrator

Subject: Request for Meteorological Data for Academic Thesis

Dear Dr. Servando,

We are third-year Bachelor of Science in Computer Science students from Laguna State Polytechnic University – Los Baños Campus (LSPU-LBC), currently conducting research for our undergraduate thesis titled *"Spatial Analysis and Multi-Source Data Fusion for Dengue Outbreak Forecasting Using an Enhanced Random Forest with Heatmap Visualization."*

To support the data requirements of our study, we respectfully request access to meteorological data from January 1, 2016 to December 31, 2021. Specifically, we are seeking weekly data per province, city, and region in the Philippines, including rainfall amount, maximum temperature, minimum temperature, wind speed, wind direction, and relative humidity. We would also appreciate any additional meteorological data relevant to weather patterns that may contribute to dengue outbreaks, such as floods and extended rainfall. Data on flood occurrences—such as dates, affected areas, and severity—would help us analyze their impact on dengue outbreaks. If available, related data on typhoons and storm surges would also be appreciated.

We assure you that all requested data will be used strictly for academic purposes, and we will maintain strict confidentiality in compliance with applicable data protection laws. Proper citation will be observed in all research outputs. We are available for further discussion at your convenience. Thank you for your time and consideration. We look forward to your favorable response.

Respectfully,

Neal Louis V. Villamor
Winchester B. Nuñez

Researchers, BS Computer Science
Email: neallouisvillamor@gmail.com
Contact: 0920 517 7866

Noted by:

CRISANTO F. GULAY
Thesis Adviser

APPENDIX C

Relevant Source Code

Data Loading and Feature Selection

```
file_path = r'C:\Users\neall\Pending Task\Final Model & System\Dataset\cleaned_dataset.csv'
df = pd.read_csv(file_path)

drop_cols = ["date", "week"]
df = df.drop(columns=[c for c in drop_cols if c in df.columns])

X = df.drop(columns=["cases"])
y = df["cases"]
```

```
tscv = TimeSeriesSplit(n_splits=5)

for i, (train_idx, test_idx) in enumerate(tscv.split(X)):
    print(f"Split {i + 1}:")
    print(f"  Train indices: {train_idx[0]} → {train_idx[-1]} ({len(train_idx)} samples)")
    print(f"  Test indices : {test_idx[0]} → {test_idx[-1]} ({len(test_idx)} samples)\n")

train_idx, test_idx = list(tscv.split(X))[-1]
X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]
y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]
```

Optimizing Model Parameters Using Bayesian Search

```
def objective_rf(trial):
    params = {
        "n_estimators": trial.suggest_int("n_estimators", 100, 1000),
        "max_depth": trial.suggest_int("max_depth", 3, 20),
        "min_samples_split": trial.suggest_int("min_samples_split", 2, 20),
        "min_samples_leaf": trial.suggest_int("min_samples_leaf", 1, 15),
        "max_features": trial.suggest_categorical("max_features", ["sqrt", "log2", None]),
        "bootstrap": trial.suggest_categorical("bootstrap", [True, False]),
    }
    model = RandomForestRegressor(random_state=42, **params, n_jobs=-1)
    score = -cross_val_score(model, X_train, y_train, cv=3,
                             scoring=make_scorer(mean_squared_error, greater_is_better=False)).mean()
    return np.sqrt(score)

study_rf = optuna.create_study(direction="minimize")
study_rf.optimize(objective_rf, n_trials=120, n_jobs=-1, show_progress_bar=False)
```

```
def objective_xgb(trial):
    params = {
        "n_estimators": trial.suggest_int("n_estimators", 100, 1000),
        "max_depth": trial.suggest_int("max_depth", 3, 20),
        "learning_rate": trial.suggest_float("learning_rate", 0.01, 0.5),
        "subsample": trial.suggest_float("subsample", 0.5, 1.0),
        "colsample_bytree": trial.suggest_float("colsample_bytree", 0.5, 1.0),
        "gamma": trial.suggest_float("gamma", 0, 10),
        "reg_alpha": trial.suggest_float("reg_alpha", 0, 5),
        "reg_lambda": trial.suggest_float("reg_lambda", 0, 5),
    }
    model = XGBRegressor(random_state=42, **params, n_jobs=1, verbosity=0)
    score = -cross_val_score(model, X_train, y_train, cv=3,
                             scoring=make_scorer(mean_squared_error, greater_is_better=False)).mean()
    return np.sqrt(score)

study_xgb = optuna.create_study(direction="minimize")
study_xgb.optimize(objective_xgb, n_trials=180, n_jobs=-1, show_progress_bar=False)
```

Training Base Models: Random Forest and XGBoost

```
rf_opt = RandomForestRegressor(random_state=42, **best_rf_params)
rf_opt.fit(X_train, y_train)
rf_opt_pred = clip_predictions(rf_opt.predict(X_test))
```

```
xgb_opt = XGBRegressor(random_state=42, **best_xgb_params)
xgb_opt.fit(X_train, y_train)
xgb_opt_pred = clip_predictions(xgb_opt.predict(X_test))
```

Ensemble Weight Optimization with Optuna

```
def objective_weights(trial):

    w_rf = trial.suggest_float("w_rf", 0.0, 1.0)
    w_xgb = 1.0 - w_rf

    rf_pred = rf_opt.predict(X_test)
    xgb_pred = xgb_opt.predict(X_test)

    combined_pred = clip_predictions((w_rf * rf_pred) + (w_xgb * xgb_pred))
    rmse = compute_rmse(y_test, combined_pred)
    return rmse

study_weights = optuna.create_study(direction="minimize")
study_weights.optimize(objective_weights, n_trials=200, n_jobs=-1, show_progress_bar=False)

best_w_rf = study_weights.best_params["w_rf"]
best_w_xgb = 1.0 - best_w_rf
```

Weighted Ensemble Construction and Prediction

```
class WeightedEnsemble:
    def __init__(self, rf_model, xgb_model, w_rf, w_xgb):
        self.rf_model = rf_model
        self.xgb_model = xgb_model
        self.w_rf = w_rf
        self.w_xgb = w_xgb

    def predict(self, X):
        rf_pred = self.rf_model.predict(X)
        xgb_pred = self.xgb_model.predict(X)
        combined_pred = (self.w_rf * rf_pred) + (self.w_xgb * xgb_pred)
        return np.maximum(0, combined_pred)

weighted_model = WeightedEnsemble(rf_opt, xgb_opt, best_w_rf, best_w_xgb)
```

APPENDIX D

Sample Input / Output

Sample Input

population	population_density	rainfall	relative_humidity	temp_max	temp_min	wind_speed	wind_direction	locality_cluster_nu	cluster_size	g_score	p_value	
250985	60	0	81.85714286	31.32857143	23.98571429	2.571428571	300	8	260	-0.1407	0.485	
250985	60	0	77.57142857	32.6	23.98571429	3.142857143	296.5714286	8	260	-0.1407	0.462	
250985	60	0	75.28571429	33.57142857	24.94285714	3.571428571	315.7142857	8	260	-0.1407	0.473	
250985	60	0	73.71428571	33.05714286	24.31428571	3	321.4285714	8	260	-0.1407	0.482	
250985	60	0	74.42857143	34.07142857	25.48571429	3.142857143	306.5714286	8	260	-0.1407	0.465	
250985	60	0	72.71428571	34.12857143	25.98571429	3.285714286	261.4285714	8	260	-0.1407	0.482	
250985	60	25.8	74	34.47142857	25.98571429	3.428571429	244.2857143	8	260	-0.1407	0.482	
250985	60	155.8	82.28571429	32.82857143	25.45714286	2.714285714	220.4285714	8	260	-0.1407	0.473	
250985	60	2	80.14285714	33.18571429	25.87142857	3.285714286	252.8571429	8	260	-0.1407	0.485	
250985	60	96.3	84.85714286	32.18571429	24.62857143	3.714285714	220	8	260	-0.1407	0.475	
250985	60	21	83	32.32857143	25.4	3.142857143	242.8571429	8	260	-0.1407	0.48	
250985	60	24	84.85714286	32.82857143	24.84285714	3	248.5714286	8	260	-0.0748	0.324	
250985	60	16.7	83.42857143	33.01428571	25.24285714	2.857142857	232.8571429	8	260	-0.1407	0.48	
250985	60	71.5	85	31.87142857	24.61428571	3	237.1428571	8	260	-0.1407	0.48	
250985	60	24.3	83.28571429	31.77142857	25.04285714	3.142857143	217.1428571	8	260	-0.0897	0.36	
250985	60	1.4	81.42857143	32.67142857	24.3	2.428571429	201.4285714	8	260	-0.1407	0.483	
250985	60	33.1	80.42857143	32.37142857	24.74285714	2.857142857	211.4285714	8	260	-0.1407	0.48	
250985	60	202.2	86.28571429	30.84285714	24.57142857	3.571428571	250	8	260	-0.1407	0.48	

Sample Output

Prediction Result

	1	2	3	4	5	6	7	8	9	10	11	12
Angeles City	88	95	99	5	3	6	8	11	17	34	30	37
Aurora	26	24	15	8	7	19	28	61	32	9	12	7
Bataan	183	200	139	31	28	33	61	111	43	45	64	66
Bulacan	341	366	356	70	51	58	200	396	329	231	227	202
Nueva Ecija	332	361	340	21	18	38	114	267	169	159	269	326
Olongapo City	26	10	4	2	1	4	4	11	5	9	15	10
Pampanga	338	363	339	94	40	50	70	111	156	255	373	352
Tarlac	285	337	262	39	29	64	119	291	265	279	425	322
Zambales	96	97	44	4	4	19	59	113	49	32	67	54

APPENDIX E

Curriculum Vitae

Neal Louis Villamayor Villamor

San Luis RD, Tuntungin Putho, Los Baños, Laguna, 4030, Philippines

09205177866 neallouisvillamor@gmail.com

Date of birth……October 07, 2003 Nationality……Filipino

PROFILE

Detail-oriented and proactive employee with extensive experience in web development. Foundational knowledge in PHP, MySQL, and Python, with an ability to design, build, and maintain functional websites with CRUD operations. Experienced in setting up computers and managing sound systems. Adept at providing practical solutions to improve workflow and efficiency. Strong problem-solving skills, capable of handling multiple tasks, and committed to delivering reliable and effective results in both individual and team projects. Continuously learning new technologies to enhance performance and contribute to organizational goals.

EMPLOYMENT HISTORY

07/2025 – 08/2025 **Department of the Interior and Local Government** – Los Baños, Laguna
Special Program for the Employment of Students

- Built the department website using Google Sites.
- Receiving documents.
- Sorting files.

07/2024 – 08/2024 **Information and Communication Systems Office** – Los Baños, Laguna
Special Program for the Employment of Students

- Setup Computer.
- Sound System Setup.

07/2023 – 08/2023 **Department of the Interior and Local Government** – Los Baños, Laguna
Special Program for the Employment of Students

- Receiving documents.
- Sorting files.

07/2022 – 08/2022 **Department of the Interior and Local Government** – Los Baños, Laguna
Special Program for the Employment of Students

- Receiving documents.
- Sorting files.

EDUCATION

2022 – 2026 **Laguna State Polytechnic University – Los Baños Campus**
Bachelor of Science in Computer Science, Major in Intelligence System

2020 – 2022 **Tuntungin Putho Integrated National High School**
Information and Communication Technology (ICT)

2016 – 2020 **Tuntungin Putho Integrated National High School**
Secondary Level / Junior High School

2010 – 2016 **Paciano Rizal Elementary School**
Primary Level / Elementary

SKILLS

Programming:
- HTML
- CSS
- JavaScript (Basics)
- PHP (Basics)
- Python (Basics)
- MySQL

Editing:
- Canva
- Capcut
- Figma

Technical Skills:
- Web Development
- Machine Learning
- Data Encoding
- MS Word
- MS PowerPoint
- MS Excel

Soft Skills:
- Problem-Solving
- Able to learn more
- Adopt Quickly
- Independent and Team-Oriented Work
- Task Prioritization and Organization

LANGUAGE

Filipino
English

PROJECT

2025 **Spatial Analysis and Multi-Source Data Fusion for Dengue Outbreak Forecasting using an Ensemble of Random Forest and XGBoost with Heatmap Visualization.**

- Developed a dengue outbreak forecasting model using Random Forest and XGBoost with integrated spatial data fusion and heatmap visualization.

Institution: Laguna State Polytechnic University – Los Baños Campus

2025 **PeriBot – Rover Room Perimeter Measurement.**

- Built an autonomous rover that measures room perimeters using sensor-based navigation.

Institution: Laguna State Polytechnic University – Los Baños Campus

2025 **Copyright Repository: A Web-Based System for Reviewing and Managing Departmental Copyright Certificates.**

- Created a web-based system for reviewing, storing, and managing departmental copyright certificates.

Institution: Laguna State Polytechnic University – Los Baños Campus

2024 **PESO Career Opportunity Application: Facilitating Job Opportunities and Skill Development**

- Developed a web application that enables users to register, explore job opportunities, and support skill development.

Institution: Laguna State Polytechnic University – Los Baños Campus

2024 **Light Control Iot System**

- Designed an IoT-based system for automating lighting functions.

Institution: Laguna State Polytechnic University – Los Baños Campus

CERTIFICATIONS

2025 **LSPU – LBC (College of Computer Studies Seminar 2025)**
Role of Technology in Sustainable Development

Winchester B. Nuñez

Los Baños, Laguna 4030 | +63 963 513 1357 | chesterbanasihan@gmail.com

PROFESSIONAL SUMMARY

A BSCS student seeking an OJT placement to apply and enhance my skills. With experience in building and designing web systems for my thesis, I am committed to learning, growing, and adding value to the workplace.

EDUCATION

Laguna State Polytechnic University **2022 - Present**

Bachelor of Science in Computer Science

Major in Intelligent Systems

Bayog Senior High School **2020-2022**

Agri-Fisherish

Los Baños Integrated School **2016 - 2020**

ACADEMIC PROJECT/THESIS

Thesis: Spatial Analysis and Multi-Source Data Fusion for Dengue Outbreak Forecasting Using An Ensemble Of Random Forest and XGBOOST With Heatmap Visualization

Institution: Laguna State Polytechnic University | 2025

Copyright Repository: A Web-Based System for Reviewing and Managing Departmental Copyright Certificates.

Developed a web-based platform for reviewing, storing, and managing departmental copyright certificates.

Institution: Laguna State Polytechnic University | 2024

Healthhub Smooth Scheduling for free medical Check-ups

A web-based system that allows users to appoint a free health check-up.

Institution: Laguna State Polytechnic University | 2024

SKILLS

Programming Languages

- HTML
- CSS
- Python (Basics)
- JavaScript (Basics)

Technical Skills

- Web development
- Web Designing
- Microsoft Word
- Machine Learing

Soft Skills

- Positive Attitude and Professionalism
- Fast Learner and Willingness to Improve
- Time Management and Task Prioritization
- Strong Work Ethic and Reliability

APPENDIX F

Publishable Paper (IEEE FORMAT)

Spatial Analysis and Multi-Source Data Fusion for Dengue Outbreak Forecasting using an Ensemble of Random Forest and XGBoost

Neal Louis V. Villamor
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***Abstract*— This study develops a forecasting approach that integrates epidemiological, demographic, meteorological, and spatial analysis to support proactive dengue management. Random Forest and XGBoost models were trained using multi-source data capturing disease patterns, climate variability, population-related risk factors and spatial analysis result. XGBoost produced stronger predictive accuracy compared to Random Forest, achieving an R-Squared of 0.926351, MAE of 4.628130, and RMSE of 7.802892. To further enhance prediction reliability, a Weighted Ensemble method was implemented with optimized weights of 0.1121 for Random Forest and 0.8879 for XGBoost. The ensemble model achieved the highest accuracy, resulting in an improved R-Squared of 0.926423, a lower MAE of 4.609461, and a reduced RMSE of 7.799064. A color-coded heatmap was used to visualize predicted dengue cases across localities and months, helping identify areas with higher outbreak risk.**

***Keywords*— Spatial Analysis, Multi-Source Data Fusion, Weighted Average Ensemble, Random Forest, XGBoost, Dengue Outbreak Forecasting**

I. INTRODUCTION

Dengue is a rapidly spreading mosquito-borne disease with no specific treatment, infecting around 390 million people annually and causing approximately 21,000 deaths worldwide [1]. Asia accounts for about 70% of the global dengue burden, with tropical and subtropical regions particularly affected [2]. The Philippines remains one of the most impacted countries, recording over 437,000 cases in 2019 and periodic outbreaks every 3–4 years [3], [4]. The primary vectors, *Aedes aegypti* and *Aedes albopictus*, transmit four dengue virus serotypes (DENV-1 to DENV-4) that cause symptoms ranging from mild fever to severe hemorrhagic dengue [5]. Disease forecasting supports early intervention by predicting outbreaks and improving healthcare preparedness [6]. Predictive models using climate and non-climate data—such as temperature, rainfall, and population density—have demonstrated strong performance in anticipating dengue epidemics. In Colombia, an early warning model detected 75% of outbreaks 1–5 months in advance [7], while similar systems in Brazil achieved up to 57% accuracy for high-risk predictions [8].

Ensemble learning, which combines multiple models, often outperforms individual models in prediction accuracy [9]. Studies show ensembles achieve lower error rates and higher reliability across applications including disease forecasting [10], [11]. Heatmaps are widely used to visualize disease distribution and risk levels, enhancing understanding and monitoring of health data [12]. Integrating heatmaps with predictive models allows for effective spatial analysis and communication of outbreak risk areas [13].

A. *Objectives of the Study*

The specific objectives of this study are:

1. To spatially analyze dengue outbreak patterns by integrating multi-source data comprising Epidemiological, Demographic, and Meteorological datasets using PostgreSQL/PostGIS, and by applying spatial clustering algorithms such as Getis-Ord Gi* and DBSCAN to identify statistically significant hotspot areas.
2. To develop weighted average ensemble that integrates Random Forest and XGBoost using the combined epidemiological, meteorological, and demographic datasets, along with spatial analysis results.
3. To evaluate the model's predictive accuracy using regression metrics (MAE, RMSE, R^2) and to visualize the forecasted dengue outbreak risks through a color-coded heatmap, to support geographically targeted public health interventions.

B. *Conceptual Framework*

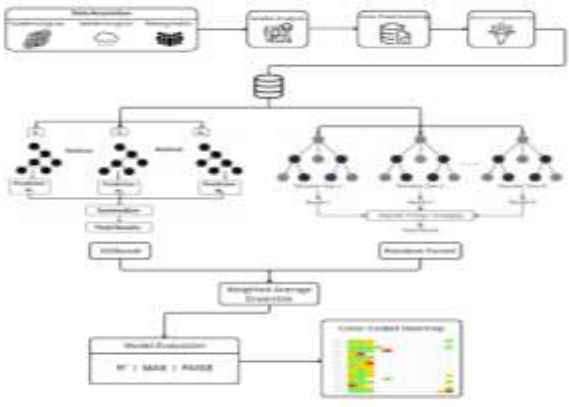


Figure 1. Conceptual Framework

As shown in Fig. 1, this study develops a Weighted Average Ensemble combining Random Forest and XGBoost to forecast dengue outbreaks in selected regions of Luzon, Philippines—specifically Region I, Region III, Region IV-A, Region V, CAR, and NCR.

The framework begins with the collection of three primary data types: epidemiological data (dengue cases), demographic data (population, population density), and meteorological data (weather conditions). These datasets undergo spatial analysis to integrate location-based attributes and identify spatial dependencies. The resulting spatial outputs are merged with the epidemiological, demographic, and meteorological datasets to form a comprehensive dataset.

Data preprocessing includes data cleaning and feature engineering to ensure model readiness and input quality. The processed data is used to train two base models—Random Forest and XGBoost. The model outputs from these base learners are then combined using a Weighted Average Ensemble approach. This integration leverages the strengths of both models, improving predictive performance and generalization.

The ensemble model's forecasting performance is evaluated using Mean Absolute Error (MAE)**,** Coefficient of Determination (R^2), and Root Mean Squared Error (RMSE)**.** The outputs include predicted dengue case counts and heatmap visualizations identifying high-risk outbreak areas across Luzon, supporting data-driven decision-making and proactive public health interventions.

II. RELATED WORKS

This study reviews key literature on spatial analysis and ensemble learning techniques for dengue outbreak prediction and control. Previous research highlights that forecasting dengue outbreaks supports early intervention and efficient public health response. In French Guiana and Cambodia, climate-based and surveillance-driven models successfully predicted most dengue outbreaks months in advance, aiding hospital preparedness [14], [15]. Integrating meteorological and long-term surveillance data has also improved prediction accuracy in Indonesia and other tropical regions [16].

Spatial analysis plays a major role in understanding dengue transmission dynamics. Geographic Information Systems (GIS) and spatiotemporal modeling have been used to identify high-risk areas and visualize disease patterns [17], [18]. Techniques such as DBSCAN clustering and Getis-Ord Gi* statistics have enhanced hotspot detection and spatial forecasting accuracy [19]. The integration of PostgreSQL and PostGIS databases has further improved spatial data processing and geospatial modeling performance [20].

Combining multiple data sources—such as epidemiological, demographic, and meteorological datasets—improves the robustness of dengue forecasting models [21]. Ensemble approaches, particularly those combining Random Forest and XGBoost, consistently outperform single models in terms of accuracy and generalization [22], [23]. Weighted average ensembles effectively leverage model strengths, reducing prediction errors in health, climate, and energy forecasting domains [24].

Color-coded heatmaps also enhance data interpretation by visualizing outbreak risks and identifying high-risk zones. They are widely used in disease surveillance systems to support targeted interventions and improve early warning systems [25].

Prior studies demonstrate the value of integrating spatial analytics, multi-source data, and ensemble learning to enhance the precision of dengue outbreak forecasting and guide data-driven public health actions.

III. METHODOLOGY

The proposed framework follows a systematic workflow comprising spatial analysis, data preprocessing, model training, and evaluation.

A. Spatial Analysis

Spatial analysis was used to identify dengue clusters and hotspots across Luzon. The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm grouped localities with similar dengue incidence patterns. Using QGIS, the parameters were set to eps = 0.45° (≈50 km) and minPts = 5 to balance cluster density and coverage. The output produced LOCALITY_CLUSTER and CLUSTER_SIZE values.

For hotspot detection, the Getis-Ord Gi* statistic was applied using Python. Data were standardized through Principal Component Analysis (PCA), and k-nearest neighbors (k = 5) ensured spatial consistency. Localities with $z \geq +1.645$ ($p < 0.10$) were identified as hotspots, and those with $z \leq -1.645$ as coldspots. The resulting z-scores and p-values were visualized through a heatmap to highlight high- and low-risk dengue areas.

B. Model Training

A weighted average ensemble combining Random Forest and XGBoost was developed to forecast dengue outbreak incidence using integrated epidemiological, demographic, and meteorological data with spatial analysis. For the Random Forest model, hyperparameters were optimized using Bayesian search in Optuna over 120 trials. The parameters included n_estimators (100–1000), max_depth (3–20), min_samples_split (2–20), min_samples_leaf (1–15), max_features (sqrt, log2, None), and bootstrap (True, False).

For XGBoost, Bayesian optimization was performed with 180 trials using the following hyperparameters: n_estimators (100–1000), max_depth (3–20), learning_rate (0.01–0.5), subsample (0.5–1.0), colsample_bytree (0.5–1.0), gamma (0–10), reg_alpha (0–5), and reg_lambda (0–5).

The ensemble model's weights were optimized across 200 trials to minimize RMSE. The Random Forest weight (w_rf) ranged from 0.0 to 1.0, while the XGBoost weight was defined as 1.0 – w_rf, ensuring balanced contributions. This optimization process produced a well-tuned ensemble capable of accurate dengue forecasting.

C. Color-Coded Heatmap Visualization

The outbreak for each month was determined by comparing predicted dengue cases against a monthly threshold derived from historical data. The threshold was calculated using the formula: threshold = mean + 2 × standard deviation. An outbreak occurred when the predicted cases for

a month exceeded this threshold, allowing identification of potential outbreak periods even without current case data.

Prediction accuracy at the locality–month level was evaluated using percentage error, computed as (actual – predicted) / actual × 100. A color-coded heatmap visualized these results, where green represented low error (≤10%), yellow indicated moderate error (≤30%), and red denoted high error (>30%). This visualization provided a clear and interpretable way to assess the model's outbreak prediction accuracy across regions and months.

D. Model Evaluation

The forecasting model's performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). These metrics measure how closely the model's predictions match actual dengue case values.

MAE represents the average absolute difference between predicted and actual values, providing a simple measure of overall prediction error. RMSE calculates the square root of the average squared differences, emphasizing larger errors and expressing results in the same units as the data. R^2 measures how much of the variance in actual dengue cases is explained by the model, indicating its overall fit. Lower MAE and RMSE values and higher R^2 scores signify better model accuracy and reliability in forecasting dengue outbreaks.

IV. RESULTS AND DISCUSSION

Spatially analyzed dengue outbreak patterns using integrated multi-source datasets and applied Getis-Ord Gi* and DBSCAN to identify statistically significant hotspot areas

A. DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

The application of the DBSCAN algorithm successfully partitioned the localities into 15 distinct clusters. Table 1 shows the results of clustering using different eps values. To identify patterns and high-risk areas of dengue cases, spatial analysis was performed using Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and hotspot detection through the Getis-Ord Gi* statistics. The latitude and longitude of all localities in the dataset were obtained using a Python script with the geopy library to automatically retrieve geographic coordinates. This process produced a complete list of localities with their corresponding spatial points. QGIS was then used to run the DBSCAN algorithm and group nearby cases together.

TABLE 1. CLUSTERING RESULTS FOR DIFFERENT EPS VALUES

eps (km)	Number of Clusters	Min Cluster Size	Max Cluster Size
10	32	260	4,420
20	28	260	4,680
30	24	260	4,940
40	22	260	5,200
50	15	260	7,540

Table 2 presents the size of each cluster. Cluster 3 is the most significant, with a size of 7,540. Following this, Cluster 6 is the next largest with a size of 780. The remaining clusters show a distinct pattern, with most having a size of 260. Clusters 5 and 14 are an exception to this, each containing 520. This cluster highlights another significant grouping of dengue cases, though smaller in extent compared to Cluster 3. Even though these groups are not as large or dense, they are still important.

TABLE 2. CLUSTER ID AND SIZE

Cluster ID	Cluster Size
Cluster 1	260
Cluster 3	7,540
Cluster 4	260
Cluster 5	520
Cluster 6	780
Cluster 7	260
Cluster 8	260
Cluster 9	260
Cluster 10	260
Cluster 11	260
Cluster 12	260
Cluster 13	260
Cluster 14	520
Cluster 15	260

*B. Getis Ord Gi**

The Getis-Ord Gi* revealed spatial clustering patterns of dengue incidence. From the dataset of 12,480, Figure 2 illustrates the results identified 146 hotspots (areas of significantly elevated incidence) and 373 coldspots (areas with significantly fewer cases than expected). The vast majority were classified as not significant, exhibiting no statistically significant clustering tendency and suggesting a random spatial distribution.

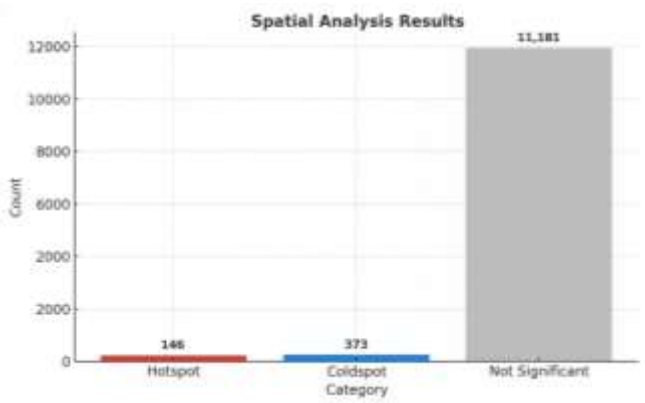


Figure 2. Results of Getis Ord Gi*

Developed Weighted Average Ensemble Model Using Random Forest and XGBoost Based on Combined Epidemiological, Meteorological, Demographic, and Spatial Analysis Inputs

A. Random Forest (Bayesian Optimization)

The Random Forest model was employed in this study to predict dengue cases using a comprehensive set of integrated inputs encompassing epidemiological, meteorological, demographic, and spatial variables. As an ensemble learning method based on multiple decision trees, Random Forest was designed to improve predictive accuracy and control overfitting by averaging the outputs of numerous trees trained on different subsets of the data. Bayesian Optimization was applied to fine-tune the model's hyperparameters, ensuring optimal depth, number of estimators, and other parameters to

maximize generalization performance. This approach enabled the Random Forest to effectively capture complex interactions among variables influencing dengue transmission, providing a stable and reliable baseline model for outbreak prediction.

TABLE 3. HYPERPARAMETER OF RANDOM FOREST

Hyperparameter	Value
n_estimators	562
max_depth	14
min_samples_split	7
min_samples_leaf	4
max_features	None
bootstrap	True

Table 3 shows the best-performing hyperparameter configuration obtained from the Random Forest optimization using Bayesian search in Optuna. The model achieved its strongest performance using 562 estimators, providing a sufficient number of trees to enhance predictive accuracy without excessive computation. The maximum depth was optimized to 14, allowing the trees to capture complex patterns while still managing overfitting. The min_samples_split value of 7 and min_samples_leaf value of 4 introduce necessary constraints on node splitting, supporting better generalization on unseen data. Also, the max_features parameter was set to None, meaning all features were considered when building each split, maximizing learning capacity. The inclusion of bootstrap=True preserves sampling with replacement, which increases tree diversity and strengthens ensemble robustness. The optimized hyperparameters in Table 3 ensure a well-tuned Random Forest model that performs reliably as a base learner in the weighted average ensemble.

B. XGBoost (Bayesian Optimization)

The XGBoost model was developed to capture the intricate nonlinear relationships between dengue cases and a diverse range of predictor variables, including climatic, demographic, and spatial factors. Known for its gradient boosting framework, XGBoost builds decision trees sequentially, where each tree corrects the errors of its predecessors, resulting in a strong predictive model. Bayesian Optimization was utilized to determine the most effective combination of hyperparameters, such as learning rate, tree depth, and regularization terms, to balance model complexity and accuracy. This process enhanced the model's ability to identify subtle, high-order dependencies among predictors, making XGBoost particularly effective in handling the heterogeneity and variability present in dengue case data.

TABLE 4. HYPERPARAMETER OF XGBOOST

Hyperparameter	Value
n_estimators	105
max_depth	4
learning_rate	0.07558836682599704
subsample	0.9792114669799131
colsample_bytree	0.5647576463467691
gamma	9.709730099421618

Table 4 presents the optimal hyperparameter configuration for the XGBoost optimization using Bayesian search in Optuna [13]. With 105 boosting rounds, the model avoids excessive complexity while retaining strong predictive performance. A maximum depth of 4 limits tree growth, reducing the likelihood of overfitting. The learning rate of approximately 0.0756 ensures gradual but stable learning from each boosting iteration. The subsample value of about 0.9792 means most of the data is used each round, balancing performance and variability during training.

The colsample_bytree value of around 0.5648 restricts the number of features used per tree, helping introduce diversity between trees and prevent reliance on specific features. A gamma value of approximately 9.71 imposes a strong constraint on further splitting, encouraging only meaningful improvements in model structure. Meanwhile, L1 (reg_alpha ≈ 2.97) and L2 (reg_lambda ≈ 4.43) regularization penalties control weight complexity, further improving generalization. With this optimized parameter set in Table 4, the XGBoost model becomes more robust and accurate, making it a strong base model for the weighted average ensemble.

C. Weighted Average Ensemble

TABLE 5. WEIGHTS OF BASE MODEL

Base Model	Weight
Random Forest (w_rf)	0.1121
XGBoost (w_xgb)	0.8879

Table 5 displays the optimal weight distribution obtained from the Optuna-based weight tuning for the weighted average ensemble model. The Random Forest model received a weight of 0.1121, indicating that it contributes a smaller portion to the final ensemble prediction. In contrast, the XGBoost model was assigned a much higher weight of 0.8879, showing that it provides the dominant influence on the ensemble's output. This weight allocation reflects the stronger predictive performance of the XGBoost model during optimization, where Root Mean Squared Error (RMSE) was used as the basis for selecting the best-performing weight combination. As shown in Table 5, these optimized weights ensure that the ensemble takes advantage of both models while prioritizing the one that delivers higher accuracy, resulting in an improved and more reliable prediction performance.

Evaluation of the Model's Predictive Accuracy Using Regression Metrics (MAE, RMSE, R^2) and Visualization of Forecasted Dengue Outbreak Risks through a Color-Coded Heatmap

A. Model Evaluation

TABLE 6. MODEL PREDICTIVE ACCURACY RESULT

Metric	Random Forest	XGBoost	Weighted Ensemble
R-Squared	0.921752	0.926351	0.926423
MAE	4.648624	4.628130	4.609461
RMSE	8.042851	7.802892	7.799064

Table 6 compares the predictive performance of the Random Forest, XGBoost, and Weighted Ensemble models using R^2, MAE, and RMSE as evaluation metrics. These values measure how closely the forecasted dengue cases match the actual recorded cases. Higher R^2 values indicate that a model explains more variation in dengue cases, while lower MAE and RMSE values indicate fewer prediction

errors. Both base models demonstrate strong forecasting performance, but each contributes differently to the overall predictive ability.

Between the two base models, XGBoost performs better than Random Forest across all metrics. However, the Weighted Ensemble—by combining both models—provides the strongest results. For R^2, the Weighted Average Ensemble improves performance by 0.004671 over Random Forest (0.926423 vs. 0.921752) and by 0.000072 over XGBoost (0.926423 vs. 0.926351). In terms of MAE, the Weighted Ensemble reduces error by 0.039163 compared to Random Forest and by 0.018669 compared to XGBoost. Meanwhile, RMSE is lowered by 0.243787 relative to Random Forest and by 0.003828 compared to XGBoost. These differences, while small in numeric scale, reflect a measurable improvement in predictive precision.

B. Color-Coded Heatmap Visualization



Figure 3. NCR

Figure 3 presents the color-coded heatmap of dengue outbreak detection across the National Capital Region (NCR). Each cell represents the predicted number of dengue cases per city and month, with colors indicating prediction confidence levels. Green cells denote highly accurate predictions that align closely with observed outbreaks. Yellow cells capture general trends with minor deviations, while red cells indicate low accuracy or potential false detections. Quezon City, Pasig City, and Taguig City show strong prediction performance, while Mandaluyong City, Navotas City, and Pasay City display less consistent results, reflecting variations in model reliability across NCR.

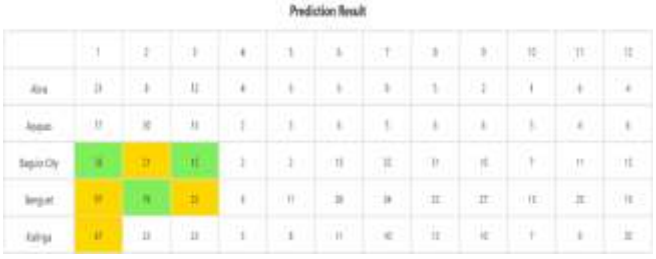


Figure 4. CAR

Figure 4 displays the color-coded heatmap for the Cordillera Administrative Region (CAR), showing monthly predicted dengue cases per province. Green cells indicate strong alignment between predicted and observed patterns, while yellow cells show moderate accuracy with slight deviations. No red cells are observed, indicating stable model performance. Baguio City achieved the highest accuracy, followed by Benguet and Kalinga, which maintained consistent moderate results. The findings suggest effective outbreak detection in areas with well-documented data, particularly in urban centers like Baguio City.

Figure 5. Region I



Figure 5 presents the heatmap for Region I (Ilocos Region), where each cell corresponds to predicted dengue cases per province and month. Green indicates high accuracy, yellow moderate, and red low accuracy. Pangasinan recorded consistently accurate predictions, while Dagupan City, Ilocos Norte, and La Union showed mixed results with several yellow and red cells. These discrepancies may be due to irregular case trends, indicating that data-rich provinces like Pangasinan support better model performance.

Figure 6. Region III



Figure 6 shows the heatmap for Region III (Central Luzon), where green marks high accuracy, yellow moderate, and red low accuracy. Bulacan, Pampanga, and Tarlac maintained strong prediction accuracy, while Nueva Ecija and Bataan displayed moderate fluctuations. Angeles City and Olongapo City recorded red cells, suggesting weaker alignment in areas with fewer cases. The results indicate that reliable predictions are achieved in provinces with comprehensive case data.

Figure 7. Region IV-A



Figure 7 presents the heatmap for Region IV-A (CALABARZON), where green indicates high accuracy, yellow moderate, and red low accuracy. Batangas, Cavite, Laguna, and Rizal consistently achieved accurate results, reflecting stable reporting and strong data availability. Lucena City displayed moderate accuracy in earlier months, while Cavite showed a few low-accuracy predictions during transitional outbreak periods. The model demonstrates strong

reliability in densely populated provinces with robust reporting systems.

Figure 8. Region V



Figure 8 shows the heatmap for Region V (Bicol Region), illustrating monthly dengue case predictions per province. Green represents high accuracy, yellow moderate, and red low accuracy. Albay and Sorsogon recorded strong prediction performance, while Camarines Norte and Catanduanes showed weaker alignment. Masbate exhibited variable accuracy across months, likely due to changing dengue case patterns in the area.

V. CONCLUSION AND RECOMMENDATIONS

This study developed a spatial and predictive framework for dengue outbreak forecasting by integrating epidemiological, demographic, and meteorological data with spatial clustering (DBSCAN and Getis-Ord Gi*). Results showed dengue cases are spatially concentrated, identifying 15 clusters and 146 significant hotspots, with Cluster 3 (NCR, CALABARZON, and Central Luzon) as the highest risk zone. The Weighted Average Ensemble model, combining Random Forest and XGBoost, achieved high accuracy (R^2 = 0.9264, MAE = 4.61, RMSE = 7.80) and outperformed the base models. The resulting heatmaps effectively visualize risk, providing a practical decision-support tool for public health planning and outbreak prevention.

Public health offices and LGUs should use predictive models to identify high-risk areas for targeted interventions and resource allocation. Future studies should integrate additional data such as mosquito counts, sanitation, and mobility to enhance prediction accuracy. Building a real-time forecasting system linked to live data (e.g., PAG-ASA, DOH) would enable timely monitoring. Exploring advanced models like deep learning could further improve forecasts, while community engagement in mosquito control and awareness remains vital. This approach can also be applied to other geographically driven health issues like malaria

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APPENDIX G

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Readability

Metrics compared to other Grammarly users

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APPENDIX H

Turnitin Result

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